

A two-stage optimization approach for aircraft hangar maintenance planning and staff assignment problems under MRO outsourcing mode

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ABSTRACT

Aircraft Maintenance Repair and Overhaul (MRO) is essential to ensure aviation safety and air transport operations. An aircraft has to temporally suspend its service and receive extensive maintenance in the hangar upon reaching the prescribed flying hours and take-off/landing times. Traditionally, each airline company undertakes hangar maintenance for its own fleet, however the nature of MRO operations has changed with the rapid development of air transport demand. Outsourcing hangar maintenance to maintenance service companies has been increasing among airlines, enabling airlines to reduce the cost of MRO while meeting the aircraft's safety requirements. After receiving the maintenance order with specifications, the maintenance service company needs to determine the maintenance schedules, parking stand allocation, aircraft movement path and staff assignment in the integrated planning problem. To integrate the abovementioned factors, a Mixed-Integer Linear Programming (MILP) model is developed. In the MILP model, geometric factors are considered, which are integrated with the multi-skill manpower assignment afterwards. Consideration of staff with multiple types of maintenance skills, aligning with the practice of sophisticated hangar maintenance, is incorporated in the model. Secondly, given the complexity of the integrated problem, a two-stage optimization approach is developed by decomposing the original model, which is coordinated by the linkage constraints between the geometric and numeric decision-making scattering in the decomposed subproblems. The results and analysis of numerical experimentation are reported, which shows: (i) the adaptability and efficiency of the two-stage optimization approach and (ii) the tractability of the two-stage optimization approach, which manage to produce good quality solutions in solving medium- to large- size instances. The impacts of maintenance demand intensity and manpower supply variation are also analysed to provide managerial insights.

1. Introduction

Aircraft Maintenance, Repair and Overhaul (MRO) are indispensable and crucial supporting activities in the aviation industry. Aircraft maintenance is strictly regulated by aviation authorities to maintain aircraft safety and airworthiness (Ng et al., 2018; Lee et al., 2018; Ng et al., 2015). An aircraft has to temporally suspend its service and receive maintenance upon meeting the prescribed flying hours and/or number of take-off/landing cycles (Ng et al., 2020). In the MRO industry, aircraft hangar maintenance involves a high operating cost, including the aircraft hangar facility, a wide range of assets as well as

manpower. From the perspective of the airlines, MRO is not value adding but is a mandatory activity in terms of airline fleet operations (Ertogral and Öztürk, 2019). The past decade has witnessed rapid expansion of the civil aviation industry, which has stimulated significant global economic growth as well as demand for commercial air transport (Lee et al., 2018; Ng et al., 2017; Ng et al., 2017; Wang et al., 2016). Given the high cost of MRO activities in aviation operations (ITAT, 2015), the severe competition in the aviation industry has stimulated airlines into reconsidering MRO practice so as to reduce their MRO costs while maintaining the fleet safety level (Eriksson and Steenhuis, 2014; Knotts, 1999). Transition in MRO practices has emerged, with

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Nomenclature

a_t	Set of maintenance requests arriving at shift t	division div , $div \in DIV$
d_t	Set of scheduled departing aircraft at shift t	a_{mt} 1, if worker m is available at shift t
A_t	Set of cumulative maintenance requests from beginning to shift t . $A_t \in \cup_{i=0}^t a_i$	S_i Set of maintenance tasks for aircraft i
D_t	Set of aircraft scheduled for departure from beginning of the planning period to shift t . $D_t \in \cup_{i=0}^t d_i$	S_m Set of maintenance tasks compatible with technician m
I	Set of incoming aircraft for maintenance services, $i \in I$	S_m^t Set of maintenance task compatible with technician m at shift t
T	Length of planning period	h_t The duration of shift t
t	Index of shift, $t \in T$	τ_{is} Required number of qualified technicians to perform maintenance task s for aircraft i
ETA_i	Estimated time of arrival of aircraft i	PD_{is} Set of predecessors before conducting task s associated with aircraft i
ETD_i	Estimated time of departure (also desired time of delivery) of aircraft i	M A sufficient large number
$MTime_i$	Aircraft i 's minimum staying time in maintenance hangar	Continuous Decision Variables
w'_{ij}	Aircraft i 's adjusted width on the occasion that aircraft j is adjacent to i	(x_i, y_i) Coordinates of reference point of aircraft i
TD_i	Aircraft i 's tail distance	Binary Decision Variables
$penalty1$	Penalty of rejecting aircraft i for maintenance service	out_{it} 1, if aircraft i is rolled out from the hangar at the beginning of shift t , and 0 otherwise
$penalty2$	Penalty of tardiness in delivering the maintenance service for aircraft i after its ETD (per minute)	in_{it} 1, if aircraft i is rolled into the hangar at the beginning of shift t , and 0 otherwise
$penalty3$	Penalty of failing to finish the maintenance tasks for aircraft i by the end of planning period	out_{iT^*} 1, if fail to finish the maintenance tasks for aircraft i by the end of planning horizon, and 0 otherwise
$Weightness_i$	Weightiness of aircraft i	p_{it} 1, if aircraft i is in hangar at shift t , and 0 otherwise
W	Hangar Width	h_{ijt} 1, if aircraft j blocks aircraft i 's pending movements at shift t , and 0 otherwise
H	Hangar Length	L_{ij} 1, if aircraft i is on the left side of aircraft j such that aircraft i is not blocked by j , and 0 otherwise
w_i	Aircraft i 's width	R_{ij} 1, if aircraft i is on the right side of aircraft j such that aircraft i is not blocked by j , and 0 otherwise
h_i	Aircraft i 's length	U_{ij} 1, if aircraft i is above aircraft j such that aircraft i is not blocked by j , and 0 otherwise
NFP_{ij}	Aircraft i and j 's NFP, which conforms with a minimum safety margin	b_{ijkt} 1, if the coordinates of aircraft j are imposed into the region of horizontal slice s_{ij}^k at shift t , and 0 otherwise
s_{ij}^k	k th horizontal slice outside the NFP_{ij}	$z_{mt, is}$ 1, if technician m is assigned to task s (belonging to aircraft i) at the beginning of shift t
$\alpha_{ij}^{kf}, \beta_{ij}^{kf}, q_{ij}^{kf}$	Parameters related to the f th edge of the horizontal slice s_{ij}^k outside the NFP_{ij}	y_{ist} 1, maintenance task s is conducted at the beginning of shift t (if minimum number of workers is met to conducted maintenance task s (on aircraft i) on shift t & precedence requirement is met)
m_{ij}	Total number of horizontal slices outside NFP_{ij}	f_{ist} 1, if working hours of maintenance task s (on aircraft i) is completed by the end of shift t
t_{ij}^k	Number of edges used to define the horizontal slices s_{ij}^k	
MPW	Set of technicians. $m \in MPW$	
MPW^{is}	Set of technicians compatible for the maintenance task s associated with aircraft i	
θ_m	Manpower cost of maintenance technician m undertaking compatible tasks for one shift	
div	Manpower planning division	
DIV	Set of manpower planning division, $div \in DIV$	
r_{is}	Required working hours by skill s to finish aircraft task s (on aircraft i)	
R_{mt}	Availability (hours) of technician m during planning	

some airline companies switching to outsourcing MRO operations for their fleets to an independent service company in order to focus on their core businesses, bringing large benefits. According to the estimation in (Marcontell, 2013), the percentage of outsourcing of MRO activities increased by around 45% from 25% between the mid-1990 s and 2012.

The aircraft hangar maintenance operation under the MRO outsourcing mode is analysed in this study, and an optimization approach for the hangar maintenance planning problem under the MRO outsourcing context is developed. In the MRO outsourcing mode, aircraft hangar maintenance demands are initiated by multiple airlines with specified requirements of the maintenance service, according to each company's internal maintenance plan for its operating fleet. Faced with increasing hangar maintenance demands, the maintenance service provider has to carry out an integrated maintenance plan aligned with multiple resources constraints. The integrated maintenance plan includes determining the service time of each incoming aircraft, the location of each incoming aircraft's parking stand as well as proper maintenance technician assignment for the maintenance tasks.

Specifically, the service time and the rolling operations for aircraft should be compatible with the multi-period parking stand allocation plans over the planning horizon. In addition, the assignment of maintenance staff should be based on the specific licenses (also known as skills) held by each technician (Chen et al., 2017), with manpower being one of critical maintenance resources (Datta et al., 2013), as each technician can only perform a particular qualified maintenance task. The licences that a technician holds also relate to the maintenance manpower cost, as a senior technician holding an advanced license usually has higher wages. Moreover, other considerations, such as team size and rest time, should be included while assigning appropriate technicians to the respective maintenance tasks. Developing such an integrated maintenance plan is challenging as there are interdependent relations among the aforementioned three core elements. The number of aircraft that a maintenance hangar can accommodate changes along the planning period as the maintenance company receives different sizes of aircraft from different airlines, and the parking stand location is not predetermined as in the conventional hangar maintenance operated

Table 1
Research works on aircraft hangar maintenance and staffing problem.

Model config.	Algorithm type	Year	Proposed methods	Benchmarking	Objective functions	Ref. No
Aircraft maintenance scheduling & reassignment	Heuristic	2003	Depth Search and random search	Exact Method ^a	Min. maintenance cost and costs incurred during the re-assignment	(Sriram and Haghani, 2003)
Aircraft rotation problem	Exact Algorithm	1997	Lagrangian relaxation and subgradient optimization	N/A	Max. benefit from making specific connections	(Clarke et al., 1997)
Aircraft parking stand allocation	Heuristic	2017	Two-step optimization	Exact Method ^a	Max. space utilization and safety margin	(Qin et al., 2017)
Line maintenance and staffing problem	Heuristic	2013	Enumerative MILP algorithm	Tabu Search	Min. manpower cost	(Belien et al., 2013)
Aircraft maintenance routing and staffing	Heuristic	2018	Leader-follower Stackelberg game; Bi-level nested ant colony optimization (ACO) algorithm	N/A	Min. expected cost of propagated delay and manpower	(Eltoukhy et al., 2018)
Aircraft maintenance routing	Heuristic	2018	Sub-route generation; route constructions	Exact Method ^a	Max. potential profit	(Eltoukhy et al., 2018)
Aircraft maintenance routing and crew scheduling	Heuristic	2014	Feasible pairing, cycle checking and postprocessing; Labelling algorithm	N/A	Max. revenue	(Diaz-Ramirez et al., 2014)
Workforce scheduling	Exact	2012	Mixed-Integer Linear Programming	N/A	Min. labor cost	(Belien et al., 2012)
Workforce scheduling	Heuristic	2015	Model enhancement	Exact Method ^a , Straight Forward Optimization	Min. labor cost	(De Bruecker et al., 2015)
Aircraft routing and maintenance staffing	Heuristic	2019	Stackelberg-Nash game model; iterative game algorithm	N/A	Min. expected propagated delay cost; Min. labor cost	(Eltoukhy et al., 2019)
Aircraft maintenance routing and tail assignment	Exact	2015	Two-stage column generation	Exact Method ^a	Min. expected propagated delay cost	(Liang et al., 2015)
Flight and maintenance planning problem	Exact	2015	Flight time availability bounding	Exact Method ^a	Max. fleet availability	(Gavranis and Kozanidis, 2015)
Fleet maintenance planning	Heuristic	2017	Cooperative game algorithm based on agent learning	N/A	Min. maintenance cost	(Feng et al., 2017)
Short-term manpower supply planning on maintenance	Heuristic	2004	Implicit enumeration	N/A	Min. total required manpower supply	(Yan et al., 2004)

^a Solved by IBM ILOG CPLEX Optimization Studio

by single airlines. In addition, due to the different arrival times, departure times and service times of incoming aircraft, the service time and the timing of the rolling in and out for each aircraft are different. Blocking of aircraft movement operations may occur in a congested period where many aircraft to be serviced arrive at the same time. In addition, inappropriate assignment of an aircraft's parking position also leads to low utilization of hangar space and unnecessary blocking of aircraft movements. Moreover, the assignment of technicians may also influence the service time of maintenance tasks, which results in changes of the service time windows and the time to complete the maintenance.

Considering the above-mentioned challenges in solving the integrated hangar maintenance planning problem, an optimization approach is developed from the perspective of the aircraft hangar maintenance company that operates under the MRO outsourcing mode. The work and methodology presented in this paper is developed and extended based on a journal and conference paper (Qin et al., 2017; Qin et al., 2019) published by us earlier. The addition of the technician assignment problem renders a more challenging optimization model than in the previous work. The hangar parking capacity, flexible parking assignment, and multi-skill technician assignment are three core difficulties in optimizing the maintenance planning. In this connection, we focus on modelling the correlations between maintenance service time scheduling, hangar layout planning and staffing. A MILP formulation is firstly developed to take in the aforementioned practical factors in hangar maintenance operations under the MRO outsourcing mode. Afterwards, a two-stage optimization approach is proposed to provide high quality solutions within a reasonable time in tackling medium- to large-scale problems. The contributions in this paper are as follows: 1) an integrated planning model incorporating the maintenance scheduling, aircraft's parking stand planning over multi-period and multi-skill technician assignment problem is developed, which aligns with the practical operations of hangar maintenance in an aircraft hangar maintenance service company operating under MRO outsourcing mode. 2) The proposed problem bridges the research gaps in the literature in studying the aircraft maintenance optimization and multi-skill technician assignment problem, with consideration of MRO outsourcing, which involves geometric factors and practical considerations in staffing. The proposed problem is an extension of the hangar planning models in the literature, fulfilling the gaps in the overall maintenance operations planning problem.

The remainder of this paper is organized as follows. A literature review is provided in Section 2, which further identifies the research gaps. The problem description, objectives and a set of constraints constituting the optimization problem is presented in Section 3. The two-stage optimization approach is presented and discussed in Section 4. The results of the numerical experiments are discussed in Section 5. Section 6 presents the conclusions and future avenues for this research.

2. Literature review

2.1. Aircraft hangar maintenance and staffing problem

Table 1 summarises the research work on the aircraft maintenance scheduling and staffing problem from different perspectives in the aviation industry. Upon meeting the prescribed flying hours and the number of take-off/landing cycles, the aircraft has to be temporarily taken out from flying service and undergoes a set of aircraft maintenance tasks (maintenance checks) to maintain the airworthiness and flying safety. The flying hours and number of take-off/landing cycles determine the specific types of maintenance checks that the aircraft needs to receive (Sriram and Haghani, 2003). Prescribed by the Federal Aviation Administration, the aircraft maintenance checks can be classified into four types, i.e. Type A, B, C and D checks, according to the intensity, duration, specification and frequency (Clarke et al., 1997). The Type D check is the most complex maintenance check, with the

most intensified maintenance and it requires significant input resources to complete the task. To complete the aircraft maintenance tasks, the relevant inputs of equipment, inventory/maintenance modules and manpower are required (Lee et al., 2008; Zhu et al., 2012; Lee et al., 2020), which involves a high operating cost. The complexity in undergoing maintenance checks was analyzed by Samaranayake, et al. (Samaranayake et al., 2002), and an engineering structure was developed to incorporate the essential elements of maintenance checks, including maintenance equipment, spare parts and modules, in order to arrange the maintenance schedule. Besides classifying the aircraft maintenance checks according to the intensity or frequencies mentioned above, it also can be classified according to where the maintenance checks are conducted. Specifically, there are two categories of maintenance checks according to the maintenance location, i.e. line maintenance and hangar maintenance (Van den Bergh et al., 2013). The aircraft maintenance check conducted at the gate or apron is categorized as line maintenance. Line maintenance is usually performed during the turnaround time, and also referred to an "on line" maintenance, as the aircraft will soon resume its flying activities (Papakostas et al., 2010). Since the Type A check involves less maintenance resource inputs and the duration of the check is short, it is classified as typical line maintenance. The other maintenance checks, i.e. B, C and D checks, involve intensive maintenance resource inputs and require relatively long maintenance lead times. Most importantly, these maintenance checks are conducted while the aircraft is taken out of its flying service in a maintenance hangar, and therefore these three types of maintenance are classified as hangar maintenance. In our previous paper (Qin et al., 2017), it was demonstrated that limited hangar space is the bottleneck in conducting increasing numbers of incoming maintenance requests from multiple airlines under the MRO outsourcing mode. In particular, we developed a static aircraft parking stand arrangement model for the maintenance company operating under the MRO outsourcing mode, which considers how the combination of aircraft of different sizes affect the hangar parking capacity in different situations. The hangar maintenance scheduling model characterizing the maintenance operations under the MRO outsourcing mode was proposed afterwards, based on the static aircraft parking stand arrangement model (Qin et al., 2019; Hu et al., 2016), which considers the blocking of aircraft movement operations in the hangar.

Traditionally, the staffing problem in aircraft maintenance work is usually incorporated into maintenance routing problem if the airline company conducts MRO operations for its own fleets (Belien et al., 2013; Eltoukhy et al., 2018; Eltoukhy et al., 2018; Diaz-Ramirez et al., 2014). For example, Chen, et al. (Chen et al., 2017) optimized a technician rostering problem in conducting aircraft hangar maintenance from the perspective of a single airline company, and it was assumed that the capacity of the maintenance hangar was constant during the planning period as the parking stand was predetermined and fixed. With the development of MRO outsourcing, it is noted that the manpower planning and workforce rostering problems are considered from maintenance service provider's perspective (Belien et al., 2013; Belien et al., 2012). For example, a workforce rostering problem is considered in (De Bruecker et al., 2015) in a line maintenance service company, which fulfills the requests for aircraft line maintenance from multiple airlines. Eltoukhy, et al. (Eltoukhy et al., 2019) developed a Stackelberg-Nash game to characterize the interdependent relations between the aircraft routing problem from the airline company and the staffing problem from aircraft maintenance service providers. In addition, the propagated delays in optimizing the aircraft maintenance routing are incorporated in the model proposed in (Liang et al., 2015). Regarding the optimization approach, fleet availability is maximized with an exact algorithm developed by Gavranis and Kozanidis (Gavranis and Kozanidis, 2015) in the maintenance scheduling problem. A cooperate game approach based on agent learning was proposed by Feng, et al. (Feng et al., 2017), which was studied based on a military aircraft unit. While optimizing the manpower planning and workforce assignment

Table 2
Research works on Multi-period layout planning and development of non-overlapping models.

Model config.	Algorithm type	Year	Proposed methods	Benchmarking	Objective functions	Ref. No
Vehicle routing with pick-ups and deliveries and two-dimensional loading constraint	Heuristic	2016	Constructive heuristic; Local search	Ant colony algorithm; multi-start biased-randomized method	Min. total travel time and required cargo handling time	(Zachariadis et al., 2016)
Capacitated vehicle routing and re-routing	Metaheuristic	2017	Improved Artificial bee colony	Conventional Artificial bee colony	Min. total travelling time	(Ng et al., 2017)
Capacitated vehicle routing with two-dimensional constraint	Metaheuristic	2015	Variable neighborhood search	Artificial bee colony	Min. cost of fleet of vehicles	(Wei et al., 2015)
Multiple pickup and delivery salesman problem with last-in-first-out constraint	Metaheuristic	2012	Simulated annealing and Ejection pool; Variable neighborhood search and probabilistic tabu search	Variable neighborhood search	Min. number of vehicle and travel distance	(Cheang et al., 2012)
Two-dimensional orthogonal packing problem with unloading constraint	Exact	2014	Branch-and-cut	Exact Method ^b	Min. total travel distance	(Cote et al., 2014)
Pickup and delivery problem with time window and LIFO loading	Metaheuristic	2015	GRASP; Local search; Adaptive crossover operator	Branch-price-and-cut	Min. handling while unloading items from vehicles	(Cherkesly et al., 2015)
Unequal area facility layout problem	Metaheuristic	2017	Hybrid genetic algorithm	GUROBI; Biased random-key genetic algorithm; Basic genetic algorithm	Min. distance and material flow between facility	(Paes et al., 2017)
Two-dimensional facility layout problem	Exact	2008	Branch-and-bound	N/A	Min. material handling cost	(Solimanpur and Jafari, 2008)
Continuous facility layout problem	Exact	2008	Branch-and-bound with permutation tree	Default branch-and-bound	Min. material handling cost	(Xie and Sahinidis, 2008)
Dynamic facility layout problem	Heuristic	2005	Evolutionary computation and dynamic programming	N/A	Min. material handling cost	(Dunker et al., 2005)
Dynamic facility problem with equal area department	Metaheuristic	2015	p-based MOPSO	Standard PSO	Min. material transportation and distance	(Xu and Song, 2015)
Fixture layout design in a multi-station assembly	Heuristic	2016	Highly optimized tolerance inspired heuristic	Basic exchange algorithm	Min. product dimensional variations	(Tyagi et al., 2016)
Cell formation with operator assignment and inter-cell layout problems	Heuristic	2014	LP-metric approach	N/A	Min. inter cell part movement, reconfiguration and training cost	(Bagheri and Bashiri, 2014)
Single and multi-floor facility layout problems	Exact	2016	Nonlinear programming	N/A	Min. interaction cost between department	(Ahmadi and Jokar, 2016)
Bidirectional circular layout problem	Exact	1996	Branch-and-bound	N/A	Min. transportation cost	(Bozer and Rim, 1996)
Cutting and packing problem for irregularly shaped pieces	Exact	2013	Branch-and-bound	Conventional branch-and-bound	Min. length of required container	(Alvarez-Valdes et al., 2013)
Nesting problems	Exact	2009	Mixed-integer programming	N/A	Min. length of required container	(Fischetti and Luzzi, 2009)
2-dimensional bin packing problems with irregular pieces and guillotine cuts	Heuristic	2015	Constructive algorithm	Earlier versions of constructive algorithm	Min. stock sheet required to cut demand pieces	(Martinez-Sykora et al., 2015)
Irregular strip packing problem	Exact	2016	Robust mixed-integer linear programming	Other relevant models of nesting problems	Min. length of required container	(Cherri et al., 2016)

^b Solved by IBM ILOG CPLEX Optimization Studio

problem in a hangar maintenance company operating under outsourcing practice, the intensity and the pattern of maintenance workload of the hangar maintenance company cannot be changed easily by requesting each airline to revise the maintenance routing decision of its own fleet, in order to alleviate or balance the workload of the maintenance company (Belien et al., 2013; Lee et al., 2019; Lee et al., 2019). As the maintenance outsourcing decisions are pre-determined by multiple airlines, the maintenance service provider aims to fulfill the maintenance demands within their permissible time windows by utilizing the available maintenance resources. Given the complexity of aircraft maintenance tasks, consideration of multiple skill types and skill levels are commonly adopted and indispensable in aircraft maintenance staffing optimization. Yan, et al. (Yan et al., 2004) considered a mathematical model for the technician assignment problem in short-term airline maintenance manpower planning, which incorporated the multi-skills of the workforce together with flexible management strategies. Chen, et al. (Chen et al., 2017) considered the multiple-skill technician assignment problem in an aircraft hangar maintenance operated by a single airline company, with a bi-objective optimization method that aims to achieve a tradeoff between minimizing the total labor cost and achieving workload allocation fairness. In the literature, most of the staff assignment and rostering problems in aircraft maintenance fall into the scope of the line maintenance operated by a maintenance company, or hangar maintenance operated by a single airline company. The research proposed in this paper aims to bridge the gaps between the multi-skill technician assignment problem in the MRO industry and hangar maintenance operation under the MRO outsourcing mode.

2.2. Multi-period layout planning and development of non-overlapping methods

Table 2 summarizes the research work on layout planning, vehicle routing with two-dimensional and loading constraints as well as nesting problems, which have direct correlations with the aircraft hangar planning addressed in this paper. To capture the characteristics of the aircraft hangar maintenance planning optimization problem under the MRO outsourcing mode, the mathematical model needs to consider the transitions of hangar parking layout along the maintenance process and entire planning period. Therefore, multi-period layout planning optimization is involved in the model development. Despite a lack of relevant literature on the aviation problem incorporating multi-period layout planning, some optimization problems in other contexts consider this factor, which share similar assumptions and problem nature. For example, the traditional Vehicle Routing Problem (VRP) can be extended by incorporating the physical shapes of goods, i.e. arranging the loading layout in a two-dimensional space, in arranging vehicle's loading layout during the simultaneous picks-up and deliveries, and such extension of VRP is classified as the composite routing and packing problem (Zachariadis et al., 2016; Ng et al., 2017). Such extension of VRP needs to optimize the two-dimensional loading/unloading layouts in the course of determining the routes of picking-up/delivering goods from suppliers to customers scattered in respective locations (Wei et al., 2015; Lee et al., 2019). While carrying out vehicle pick-up/delivery routes, imposing Last-In-First-Out (LIFO) loading/unloading constraints is commonly adopted in the mathematical model, to alleviate the negative effects of retrieving the goods loaded in the inner area of the vehicle that is blocked by the other goods loaded in the outside area (Zachariadis et al., 2016; Wei et al., 2015; Cheang et al., 2012; Cote et al., 2014; Cherklesly et al., 2015). In addition, another typical multi-period layout planning problem is the Facility Layout Problem (FLP), which is commonly studied in the context of the construction industry where the facilities have to be arranged at different locations within the construction site during the project period. Specifically, the facilities are modelled as rectangles of different sizes in a two-dimensional space, and the materials delivery cost in the facilities is minimized in the

common objective function setting (Paes et al., 2017; Solimanpur and Jafari, 2008; Xie and Sahinidis, 2008; Anjos and Vieira, 2017). The static Facility Layout Problem can be extended to the dynamic FLP, which arranges and changes the facilities within the construction site during the planning period. The common objective under dynamic FLP is to minimize the cost of changing facility locations at different stages of the project period (Dunker et al., 2005; Xu and Song, 2015). Despite the extensive research in the literature studying the layout planning problem (Tyagi et al., 2016; Mohammadi and Forghani, 2014; Bagheri and Bashiri, 2014; Ahmadi and Jokar, 2016; Bozer and Rim, 1996), the available methods and approaches are not applicable to be directly applied to the hangar layout planning in maintenance hangar. In particular, the work reported in the literature usually assumes that the facilities are modelled in rectangular shapes, while modelling aircraft in rectangular shapes leads to a great loss in hangar space utilization. In addition, the LIFO constraint imposed on the extended VRP can be regarded as a soft constraint in the optimization of hangar maintenance planning, considering the flexible maintenance service window for each aircraft. Furthermore, due to the single entrance/exit in an aircraft hangar and the physical shape of aircraft, blocking during aircraft movement operations has significant negative impact on the maintenance efficiency, which needs to be carefully modelled and optimized in hangar maintenance planning.

The problem of arranging aircraft parking positions is a version of the cutting and packing problem (also known as the nesting problem in the literature) in a two-dimensional space. Given that the irregular shape of aircraft needs to be modelled to reduce the wastage of hangar space, the approach to prevent irregular shapes from overlapping should be adopted. In the literature, one of the most widely adopted methods to examine whether the polygons overlap with each other is the No-Fit Polygon (NFP) in the nesting problem. A tutorial provided by Bennell and Oliveira (Bennell and Oliveira, 2008) and Bennell and Oliveira (Bennell and Oliveira, 2009) described a method to obtain NFP for a pair of polygons, which is applicable for generating NFP between aircraft, since aircraft are classified as non-convex polygons. To model the NFP in the MILP formulation, a horizontal slice modelling method was introduced in (Alvarez-Valdes et al., 2013), which is an enhancement of the model developed in (Fischetti and Luzzi, 2009). Moreover, the horizontal slicing method is further used to solve the nesting problem with guillotine cuts in (Martinez-Sykora et al., 2015). Regarding the latest development of the mathematical model incorporating NFP, two MILP formulations developed by Cherri, et al. (Cherri et al., 2016) decomposed non-convex polygons to obtain multiple convex polygons representing the original shape which could then generate NFP. The computational results demonstrate that the newly developed model is robust in tackling non-convex polygons of various shapes.

3. Problem description, assumptions and model development

3.1. Problem description

Considering an aircraft is temporarily taken out of service upon meeting the prescribed flying hours and number of take-off/landing cycles (Van den Bergh et al., 2013), the aircraft has to undergo hangar maintenance, and airline companies have adopted the MRO outsourcing practice. Airline companies send the aircraft in their fleet to an aircraft hangar maintenance company for relevant maintenance servicing according to each airline's internal maintenance routing plan for the fleet. After receiving the multiple hangar maintenance requests from airline companies, an integrated aircraft hangar maintenance plan has to make the following decisions to fulfill the incoming service requests:

- a service time schedule specifying the maintenance service period of each aircraft, including the rolling in/out timing of each incoming aircraft;

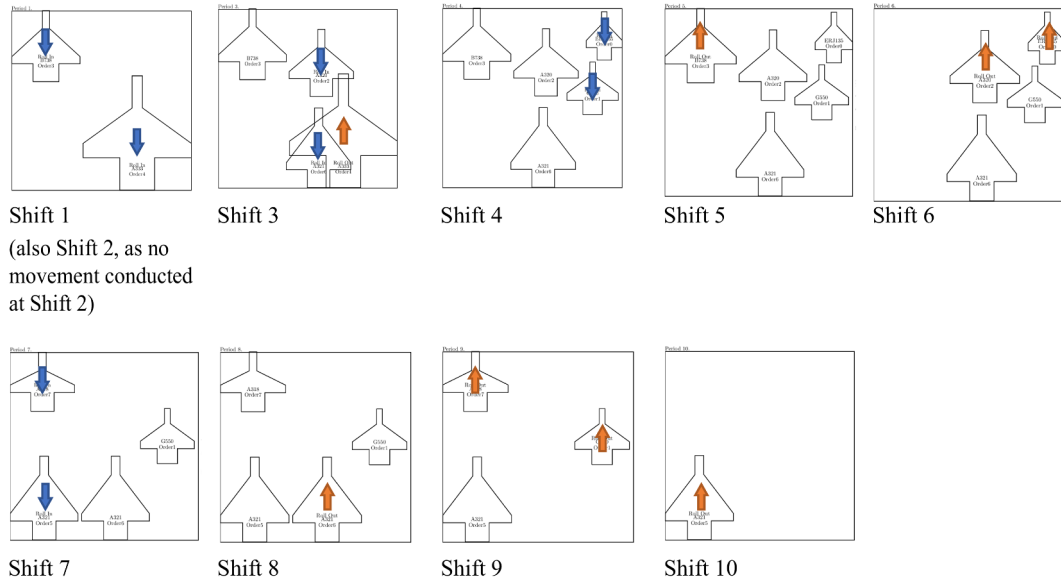


Fig. 1. Transitions of hangar layouts in maintenance planning.

- multi-period hangar layout arrangements through the entire planning period, which aligns with the maintenance schedule. The hangar parking layouts specify the movement operations of all aircraft that lead to the changes of the hangar layouts;
- Staff assignment to each maintenance tasks associated with the incoming aircraft for maintenance service.

The objective of the optimization problem is to minimize the sum of 1) the penalty due to tardiness in completing the maintenance service; 2) the penalty of rejecting a maintenance request; 3) the penalty cost of failure to complete the maintenance; and 4) the manpower costs associated with the maintenance activities. The transition of hangar layout plans are illustrated in Fig. 1, which specifies the parking positions of the aircraft along the planning period. The continuity of the hangar layout is ensured by examining the positions of aircraft in the present shift, last shift and coming shift.

3.2. Non-overlapping methods and enhancement of hangar space utilization

To fully utilize the limited hangar space for accommodating incoming aircraft from multiple airlines, the modelling of aircraft in the

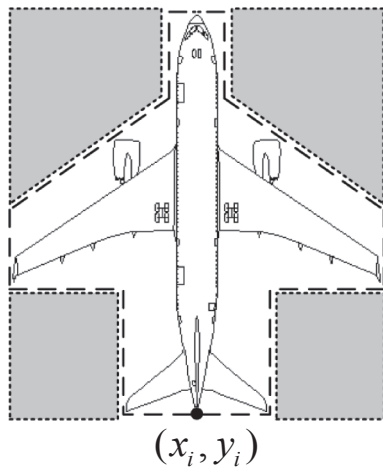


Fig. 2. Modelling of aircraft and coordinates point.

mathematical model should be accurate and close to their physical shape. In the relevant literature, aircraft are modelled as rectangles in the parking problem (Zheng et al., 2019), which induces wastage of hangar space as the aircraft is not modelled close to its physical shape. In this connection, to make the most of the limited hangar space, an aircraft is modelled as a non-convex polygon, as shown in Fig. 2. Fig. 2 illustrates the difference between modelling aircraft as rectangles and as non-convex polygons, and the shadow area indicates the area that can be utilized for parking aircraft in the hangar when modelling aircraft as non-convex polygons. To prevent two aircraft from overlapping while modelling them as non-convex polygons, the linear constraints characterizing the non-overlapping requirements become more complex, compared with modelling as rectangles. In this connection, the No-Fit Polygon is adopted to generate the non-overlapping constraints information in this model to prevent aircraft from overlapping, an approach which is commonly used in the nesting problem for packing irregular shaped items. To generate a No-Fit Polygon, we first denote the centre of polygon's bottom in two-dimensional space as the reference point with the coordinates (x_i, y_i) to determine aircraft p_i 's position in the maintenance hangar, as shown in Fig. 2. The non-overlapping constraints are derived accordingly from the non-convex shape of the aircraft. The basic concept of obtaining the Non-Fit Polygon (NFP) between two polygons is as follows: given two polygons p_i and p_j in Fig. 3, p_i remains relative stationary, and p_j is the relative movable one. The NFP_{ij} for this pair of polygons is obtained by recording the movement path of p_j 's reference point (i.e. bottom left point on polygon p_j) when p_j slides along p_i 's shape, keeping two polygons touching each other during the movement but never overlapping. In this connection, NFP_{ij} indicates the forbidden region of the reference point of p_i .

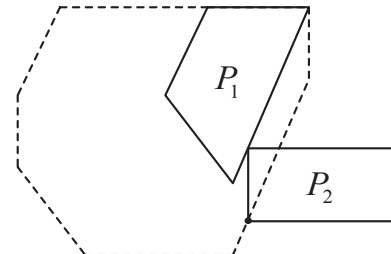


Fig. 3. NFP between two polygons.

To prevent two polygons from overlapping, the reference point of p_i cannot be located inside NFP_{ij} or its boundary, otherwise p_i would overlap or collide with p_j . The feasible regions for locating the reference point of p_j would be any location outside NFP_{ij} . Therefore, to prevent two aircraft from overlapping, the constraint should ensure that the coordinates of the movable aircraft are always outside the NFP region.

To incorporate the geometric information of NFP between a pair of polygons (aircraft) into the mathematical model, Alvarez-Valdes, et al. (Alvarez-Valdes et al., 2013) developed a horizontal slicing method to convert the geometric information to parameters in developing the model. In particular, the region outside the NFP is partitioned by multiple horizontal lines, which results in a multiple horizontal slice region outside the NFP. Note that each horizontal slice is obtained by drawing one or two horizontal line(s) outwards from each vertex of the NFP. Afterwards, the horizontal slice is characterised and defined by the edge (boundary) of the NFP associated with the slice, and the horizontal line(s) (Fig. 4(a)). After obtaining the horizontal slices, a set of binary decision variables b_{ijk} is defined, and each binary variable in this set is associated with one horizontal slice defined by the horizontal slicing method mentioned above. If $b_{ijk} = 1$, the reference point of the relative movable polygon p_j is placed in the slice k .

To convert the geometrical information of the NFP into the mathematical model, converting the slices information outside the NFP into a linear equation is necessary. Several linear equations are required to characterise one slice outside the NFP. To define each linear equation, we take two points of the segment (usually two vertexes) to calculate the slope of the segment. After defining the slope with the coordinates of the vertex, the associated linear equation can be derived and converted to a general form $\alpha_{ij}^{kf}x + \beta_{ij}^{kf}y = q_{ijk}$. To apply the linear equation into the non-overlapping linear constraint between two aircraft, we denote α_{ij}^{kf} , β_{ij}^{kf} and q_{ijk} as parameters defining the f th linear segment of the k th slice associated with NFP_{ij} , and $\alpha_{ij}^{kf}(x_j - x_i) + \beta_{ij}^{kf}(y_j - y_i) = q_{ijk}$ is the expression of the equation, where (x_i, y_i) , (x_j, y_j) are the coordinates of aircraft i and j . m_{ij} is the total number of horizontal slices. The following expression is a general form of the constraint preventing overlap:

$$\alpha_{ij}^{kf}(x_j - x_i) + \beta_{ij}^{kf}(y_j - y_i) \leq q_{ijk} + M \cdot (1 - b_{ijk}), \quad \forall i, j \in P, i \neq j, k = 1, 2, \dots, m_{ij}$$

While arranging an aircraft's parking position, it is not allowed to have each pair of aircraft touch each other, and a minimum safety distance must be imposed in the real situation. To impose such minimum safety distance (also known as safety margin), one approach is to adjust original NFP's boundary by moving its edges outwards at a distance equal to the safety margin n (Fig. 4(b)). In this connection, the coordinates of the relative movable aircraft in that pair is restricted by the revised NFP, and the shortest distance from any direction between aircraft is ensured. The safety margin is prescribed as one metre between two aircraft in this model.

3.3. MILP model

3.3.1. Assumptions in service time scheduling and layout planning

The assumptions in modelling service time scheduling and layout planning are presented as follows:

- The parameters related to the timing of the maintenance request are deterministic, including the estimated time of arrival (ETA), estimated time of departure (i.e. desirable delivery/finish time), and the maintenance time for each maintenance tasks associated with the aircraft. The duration of the aircraft's movement operations (roll in & out) are negligible. Unexpected or unscheduled maintenance requests are not considered;
- An aircraft's parking stand position cannot be changed during its service period. The movement path is a straight line, and turning the

direction of the movement path is not allowed;

- An aircraft is not allowed to leave the hangar without finishing all the maintenance tasks. The penalty cost of failure to deliver is imposed if the maintenance tasks cannot be finished in the planning horizon. After leaving the hangar, revisiting is not allowed for all aircraft;
- If the movement operations, i.e. rolling in & out, are scheduled to be conducted at shift t , then its movement path cannot have any obstacles created by the other aircraft in the hangar. Otherwise, the schedule, or the parking stand position for the arrival aircraft, have to be revised due to blockage;

3.3.2. Assumptions in manpower planning

- For each incoming aircraft, the complete maintenance request is broken-down into a series of maintenance tasks with precedence relations.
- Each maintenance task requires one or more types of maintenance skills, which correlates to the licenses held by the technicians, as shown in Fig. 5.
- On each shift, the exact numbers of required technicians are assigned if the maintenance task is scheduled to be conducted in that shift.
- Senior maintenance technicians are allowed to conduct maintenance tasks requiring junior-level skills.
- If the precedence relations between two maintenance tasks on one aircraft are imposed, then the later task cannot be conducted before finishing the earlier one.
- The assignments of maintenance technicians conform with the resting time requirement, i.e. no two consecutive shifts are allowed.

3.3.3. Notations, parameters and decision variables

The parameters of hangar maintenance planning include:

- **Maintenance demand information:** The information on incoming aircraft for hangar maintenance, including aircraft type, breakdown of maintenance checks with the specifications of maintenance skills and the size of the maintenance team. Each maintenance request has its own weightiness (importance level) contributing to the objective function, desired service window, including the estimated time of arrival (ETA), and the desired estimated time of departure (ETD). Delivery after ETD induces a tardiness cost.
- **Non-overlapping constraint's parameters:** The necessary geometric information is related to aircraft dimensions, the No-Fit Polygons for generating non-overlapping constraints, and the dimensions of the maintenance hangar.
- **Manpower information:** Information on Multi-skill maintenance technicians, with licenses held by respective persons, and the available working time in a particular subsection and the manpower cost is required.

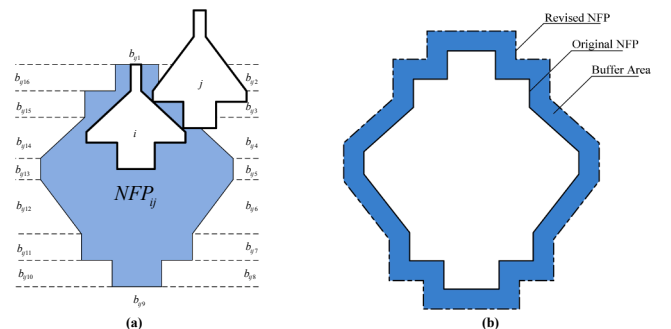


Fig. 4. Horizontal slices outside NFP.

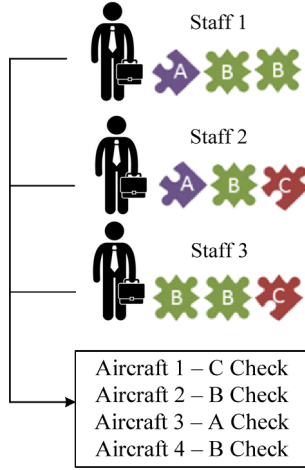


Fig. 5. Multi-skill maintenance technician assignment.

3.3.4. Objective and constraints

$$\begin{aligned} \min \sum_{i \in A_T} \text{Weightness}_i & \left[\left(1 - \sum_{t \geq \text{ETA}_i} \text{in}_{it} \right) \cdot \text{penalty}_1 \right. \\ & + \sum_{t \geq \text{ETD}_i} \text{out}_{it} (t - \text{ETD}_i) \cdot \text{penalty}_2 + \text{out}_{iT^*} \cdot \text{penalty}_3 \Big] \\ & + \sum_{m \in M} \sum_{t \in T} \sum_{s \in S_m} z_{ms, it} \cdot \theta_m \end{aligned}$$

The sum of the three penalty costs as well as the manpower cost in servicing the incoming aircraft for hangar maintenance from multiple airlines is minimized in the objective function, which can be broken down into: 1) the penalty costs of tardiness while delivering the maintenance requests; 2) penalty costs of failing to finish and deliver the maintenance request in the planning period; 3) penalty costs of the profit loss cost in rejecting a maintenance request; and 4) the manpower cost in conducting maintenance tasks for an aircraft.

The maintenance planning timeline is indexed by the shift through the planning period (Fig. 6), which is different from our previous work in (Qin et al., 2019) where the timeline is indexed by the possible movement timings of the aircraft. Each time point on the timeline represents each shift t in this model. The decisions to be made at shift t involve the movement operations, the parking position, manpower assignments as well as the maintenance task status (whether the task is in progress or finished). The position decision variables are not indexed by the shift as the position remains unchanged once rolled in. The other auxiliary decision variables determine the positions relation between each pair of aircraft, as well as movement operations, i.e. out_{iT^*} , p_{it} , h_{ijt} , L_{ij} , R_{ij} , U_{ij} and b_{ijkt} , are indexed by shift t to establish the continuity through multiple shifts.

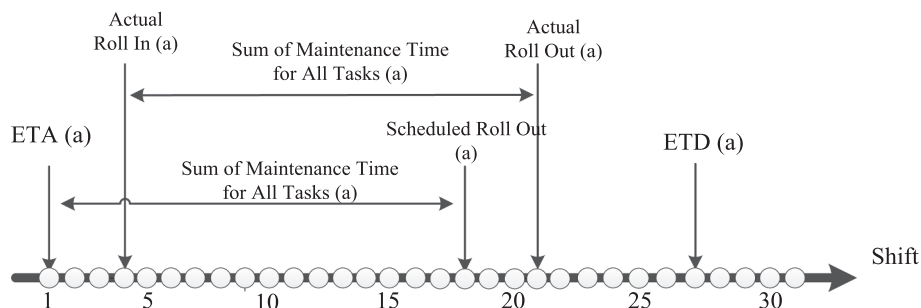


Fig. 6. Planning Horizon indexed by shifts.

For better presentation of the mathematical formulation, the constraints are divided into several subsections according to the functions. $s. t.$

1) Geometric constraints to prevent overlapping

The aircraft have to be parked in the maintenance hangar's boundary, and each pair of aircraft to be parked should not overlap with each other and be separated by the minimum safety margin.

$$x_i + w_i/2 \leq W, \quad \forall i \in I \quad (1)$$

$$x_i \geq w_i/2, \quad \forall i \in I \quad (2)$$

$$\begin{aligned} y_i + h_i & \leq H, \quad \forall i \in I \\ \alpha_{ij}^{kf} (x_j - x_i) + \beta_{ij}^{kf} (x_j - x_i) & \leq q_{ij}^{kf} + M \cdot (1 - b_{ijkt}), \\ \forall i, j \in A_t, \quad \forall k = 1, 2, \dots, m_{ij}, \quad \forall f = 1, 2, \dots, t_{ij}^k, \quad \forall t \geq 0 \end{aligned} \quad (3)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \leq p_{it}, \quad \forall i, j \in A_t, \quad \forall t \geq 0 \quad (5)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \leq p_{jt}, \quad \forall i, j \in A_t, \quad \forall t \geq 0 \quad (6)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \leq 1 - \text{out}_{it}, \quad \forall i \in D_t, \quad \forall t \geq 0 \quad (7)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \leq 1 - \text{out}_{jt}, \quad \forall j \in D_t, \quad \forall t \geq 0 \quad (8)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \geq p_{it} + p_{jt} - 1, \quad \forall i, j \in A_t \setminus D_t, \quad \forall t \geq 0 \quad (9)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \geq p_{it} + p_{jt} - (\text{out}_{it} + \text{out}_{jt}) - 1, \quad \forall i, j \in D_t, \quad \forall t \geq 0 \quad (10)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \geq p_{it} + p_{jt} - \text{out}_{it} - 1, \quad \forall i \in D_t, \quad \forall j \in A_t \setminus D_t, \quad \forall t \geq 0 \quad (11)$$

$$\sum_{k=1}^{m_{ij}} b_{ijkt} \geq p_{it} + p_{jt} - \text{out}_{jt} - 1, \quad \forall i \in A_t \setminus D_t, \quad \forall j \in D_t, \quad \forall t \geq 0 \quad (12)$$

The constraint sets (1) – (3) ensure that the entire aircraft should be parked within the hangar boundary. The No-Fit Polygon (NFP) geometric information discussed in Section 3.2 is converted to the horizontal slicing formulation form expressed in Constraint (4) in Section 3.2. The whole set of constraints preventing aircraft parked in the hangar from overlapping is presented in constraints (4) – (12). To activate the non-overlapping constraint between two aircraft, the two aircraft must be parking in the hangar in the same shift t (Constraints (9) – (12)), otherwise the relevant constraints are deactivated in other scenarios, e.g. any one of the aircraft in that pair is not scheduled to

park in the shift, or the aircraft is leaving the hangar in the same shift (Constraints (5) – (8)). As discussed in Section 3.2, the binary variables b_{ijkt} set is used to denote multiple horizontal slices around the region of NFP_{ij} . If the non-overlapping constraint (Constraint (4)) is activated, one of b_{ijkt} in that set must to take the value one.

2) Blocking during aircraft movement operations

Aircraft movement operations can be conducted under the condition that the other aircraft in the hangar do not block the movement path of the aircraft waiting to move. To determine whether the aircraft waiting to move is blocked by the other aircraft, four auxiliary binary decision variables h_{ijt} , L_{ij} , R_{ij} , U_{ij} assist in determining the position relations between the aircraft waiting to move and the aircraft already in the hangar. If any blocking of aircraft movement path exists, the movement operations have to be postponed or revised.

$$(x_i + w'_{ij}/2) - (x_j - w'_{ij}/2) \leq M \cdot (1 - L_{ij}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (13)$$

$$(x_i - w'_{ij}/2) - (x_j + w'_{ij}/2) \geq -M \cdot (1 - R_{ij}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (14)$$

$$(y_i + TD_i) - (y_j + TD_j) \geq -M \cdot (1 - U_{ij}) \quad \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (15)$$

$$(1 - h_{ijt}) \geq \frac{1}{6} \cdot [L_{ij} + R_{ij} + U_{ij} + in_{jt} + out_{jt} + (1 - p_{jt})] \quad \forall i \in A_t, \forall j \in D_t, \forall t \geq 0 \quad (16)$$

$$(1 - h_{ijt}) \leq L_{ij} + R_{ij} + U_{ij} + in_{jt} + out_{jt} + (1 - p_{jt}) \quad \forall i \in A_t, \forall j \in D_t, \forall t \geq 0 \quad (17)$$

$$(1 - h_{ijt}) \geq \frac{1}{5} \cdot [L_{ij} + R_{ij} + U_{ij} + in_{jt} + (1 - p_{jt})] \quad \forall i \in A_t, \forall j \in A_t \setminus D_t, \forall t \geq 0 \quad (18)$$

$$(1 - h_{ijt}) \leq L_{ij} + R_{ij} + U_{ij} + in_{jt} + (1 - p_{jt}) \quad \forall i \in A_t, \forall j \in A_t \setminus D_t, \forall t \geq 0 \quad (19)$$

The relations between the aircraft waiting to move and the other aircraft already parked in the hangar are characterized in constraint sets (13) – (19). Three auxiliary binary variables (L_{ij} , R_{ij} and U_{ij}) indicate the position relation between the aircraft i waiting to move and the other aircraft j . If the auxiliary binary variables take the value 1, it means that the aircraft i waiting to move is on the left side, right side or upper position of the aircraft j already parked in the hangar, respectively, and its movement path is not blocked by aircraft j .

To indicate if the aircraft waiting to move is blocked by the other aircraft parked in the hangar in shift t , the movement path status is reflected in the value of the binary variable h_{ijt} and its value is coordinated by constraint sets (16) – (19). If none of the values among L_{ij} , R_{ij} and U_{ij} has a value of one, it means that the movement path of the aircraft i is blocked by aircraft j , and under such a situation h_{ijt} takes the value one. Several exceptions exist such that the movement path is clear even if the value of h_{ijt} equals one: 1) aircraft j is also waiting to move in the same shift t , or 2) aircraft j is not in the hangar at shift t .

If the aircraft i waiting to move is blocked by any one of the other aircraft j parked in the hangar, the movement operations for aircraft i is delayed (Constraints (20) – (21)). The movement operations have to be suspended until the movement path is cleared for the departing aircraft. The parking position for the arriving aircraft can be adjusted, or the rolling in time suspended until the movement path is clear.

$$out_{it} \leq 1 - \frac{1}{|A_t \setminus i|} \cdot \sum_{j \in A_t \setminus i} h_{ijt}, \quad \forall i \in D_t, \forall t \geq 0 \quad (20)$$

$$in_{it} \leq 1 - \frac{1}{|A_t \setminus i|} \cdot \sum_{j \in A_t \setminus i} h_{ijt}, \quad \forall i \in A_t, \forall t \geq 0 \quad (21)$$

3) Duration of stay in the hangar

The duration of each aircraft stay in the hangar must be long enough to complete its maintenance tasks, and the minimum stay time duration of the aircraft should be the sum of the lead times of all maintenance tasks associated the aircraft, which is denoted as $MTime_i$.

$$\left(\sum_{t \geq ETD_i} out_{it} \cdot t - \sum_{t \geq ETA_i} in_{it} \cdot t \right) + M \cdot \left(1 - \sum_{t \geq ETA_i} in_{it} \right) + M \cdot \left(1 - \sum_{t \geq ETD_i} out_{it} \right) \geq MTime_i, \quad \forall i \in I \quad (22)$$

$$p_{it} = \sum_{ETA_i \leq m \leq t} in_{im}, \quad \forall i \in A_T, \forall ETA_i \leq t \leq ETD_i \quad (23)$$

$$p_{it} = \sum_{ETA_i \leq m \leq t} in_{im} - \sum_{ETD_i \leq m \leq t-1} out_{im}, \quad \forall i \in A_T, \forall t \geq ETD_i + 1 \quad (24)$$

$$\sum_{t \geq ETA_i} in_{it} \leq 1, \quad \forall i \in I \quad (25)$$

$$\sum_{t \geq ETD_i} out_{it} \leq 1, \quad \forall i \in I \quad (26)$$

$$out_{it} \leq \sum_{ETA_i \leq m < t} in_{im}, \quad \forall i \in I, \forall t \geq ETD_i \quad (27)$$

$$(1 - out_{iT*}) \leq \sum_{t \geq ETD_i} out_{it} + M \cdot \left(1 - \sum_{t \geq ETA_i} in_{it} \right), \quad \forall i \in I \quad (28)$$

$$(1 - out_{iT*}) \leq \sum_{t \geq ETD_i} out_{it} + M \cdot \left(1 - \sum_{t \geq ETA_i} in_{it} \right), \quad \forall i \in A_T \quad (29)$$

Constraint (22) ensures that the duration of each aircraft stay in the maintenance hangar has to be longer than the minimum maintenance time, which is equal to the sum of the lead times of all the maintenance tasks associated with the aircraft. Constraint (22) is relaxed, given the condition that the aircraft is not accepted for maintenance service, or the maintenance tasks cannot be finished within the planning horizon.

Binary variable p_{it} indicates whether aircraft i is parked in the hangar or not, and Constraints (23) – (24) ensure that the binary variable takes an appropriate value. p_{it} takes the value one along the shifts from the time that aircraft i moves into the hangar, until the time it leaves the hangar. Constraint (25) prescribes that the aircraft's rolling in operations can be conducted on or after its estimated time of arrival (ETA), and the same logic applies to Constraint (26) prescribing the roll out operations' decision-making. Constraint (27) makes sure the roll out operations occurs after the aircraft has rolled into the hangar. If the maintenance tasks of aircraft i cannot be finished in the planning horizon, the aircraft maintenance company fails to deliver the maintenance request and the binary variable out_{iT*} takes the value one by Constraints (28) – (29).

4) Range of decision variables in maintenance scheduling and layout planning

$$x_i, y_i \geq 0 \quad \forall i \in I \quad (30)$$

$$b_{ijkt} \in \{0, 1\} \quad \forall i, j \in A_t, k = 1, 2, \dots, m_{ij}, \forall t \geq 0 \quad (31)$$

$$p_{it} \in \{0, 1\} \quad \forall i \in A_t, \forall t \geq 0 \quad (32)$$

$$in_{it} \in \{0, 1\}, \quad \forall i \in A_t, \forall t \geq 0 \quad (33)$$

$$out_{it} \in \{0, 1\}, \forall i \in D_t, \forall t \geq 0 \quad (34)$$

$$h_{ijt}, L_{ij}, R_{ij}, U_{ij} \in \{0, 1\} \forall i \in A_t, \forall j \in A_t \setminus i, \forall t \geq 0 \quad (35)$$

The coordinates of aircraft parking in the hangar take positive values (Constraint (30)). The binary variables related to the maintenance scheduling and layout planning optimization section are denoted in constraint sets (31) – (35).

$$L_{ij} + L_{ji} \leq 1, \forall i, j \in I, j \neq i \quad (36)$$

$$R_{ij} + R_{ji} \leq 1, \forall i, j \in I, j \neq i \quad (37)$$

$$L_{ij} \leq R_{ji}, \forall i, j \in I, j \neq i \quad (38)$$

$$R_{ij} \leq L_{ji}, \forall i, j \in I, j \neq i \quad (39)$$

Constraint sets (36) – (39) tighten the geometric relations between each pair of aircraft while deciding on its relative position. For example, only one binary variable in set $\{L_{ij}, L_{ji}\}$, and set $\{R_{ij}, R_{ji}\}$ can take the value one, which is imposed by Constraints (36) – (37). Moreover, similar logic applies to Constraints (38) – (39). For example, if the value of R_{ji} equals zero, the value of L_{ij} cannot take the value 1, for consistency.

5) Staff Assignment Components:

$$y_{ist} \leq p_{it}, \forall i \in I_t, \forall s \in S_i, t \geq ETA_i \quad (40)$$

$$y_{ist} \leq \frac{1}{|PD_s|} \cdot \sum_{s' \in PD_{is}} f_{s'd}, \forall s \in S, t \geq ETA_i \quad (41)$$

$$y_{ist} \leq \frac{1}{\tau_{is}} \cdot \sum_{m \in MPW^{is}} z_{mt, is}, \forall i \in I, s \in S_i, t \geq ETA_i \quad (42)$$

$$y_{ist} \geq \frac{1}{\tau_{is}} \cdot \sum_{m \in MPW^{is}} z_{mt, is}, \forall i \in I, s \in S_i, t \geq ETA_i \quad (43)$$

$$\sum_{m \in MPW^{is}} z_{mt, is} \leq \tau_{is}, \forall i \in I, s \in S_i, t \geq ETA_i \quad (44)$$

$$\sum_{m \in MPW^{is}} z_{mt, is} \leq \tau_{is} \cdot (1 - f_{ist}), \forall i \in I, s \in S_i, t \geq ETA_i \quad (45)$$

$$z_{mt, is} \leq p_{it}, \forall m \in MPW^{is}, \forall i \in I_t, \forall s \in S_i, t \geq ETA_i \quad (46)$$

$$M \cdot f_{ist} \geq \sum_{t' > t \geq ETA_i} y_{ist'} \cdot h_{t'} - r_{is}, \forall i \in I, \forall s \in S_i, t \geq ETA_i \quad (47)$$

$$-M \cdot (1 - f_{ist}) \leq \sum_{t' > t \geq ETA_i} y_{ist'} \cdot h_{t'} - r_{is}, \forall s \in S_i, \forall i \in I, t \geq ETA_i \quad (48)$$

$$\sum_{t \geq ETA_i} y_{ist} \cdot h_t \geq r_{is} - M \cdot \left(1 - \sum_{t \geq ETA_i} in_{it}\right), i \in I, s \in S_i \quad (49)$$

$$\sum_{s \in S_m^I, i \in I} z_{mt, is} \leq a_{mt}, \forall m \in MPW, \forall t \in T \quad (50)$$

$$\sum_{t' \in Div_d, t > t'} \sum_{i \in I} \sum_{s \in S_m^I} z_{mt', is} \cdot h_{t'} \leq R_{mdiv}, \forall m \in MPW, \forall div \in DIV \quad (51)$$

$$\sum_{s \in S_m^I} z_{mt, is} + \sum_{s \in S_{m+1}^I} z_{m(t+1), is} \leq 1, \forall m \in MPW, \forall t \in T \quad (52)$$

$$out_{it} \leq \frac{1}{|S_i|} \cdot \sum_{s \in S_i} f_{ist}, \forall i \in I, t \geq ETD_i \quad (53)$$

$$z_{mt, is} \in \{0, 1\}, \forall m \in MPW, \forall t \in T, \forall i \in I, \forall s \in S_i \quad (54)$$

$$y_{ist} \in \{0, 1\}, \forall i \in I, \forall s \in S_i, t \geq ETA_i \quad (55)$$

$$f_{ist} \in \{0, 1\}, \forall i \in I, \forall s \in S_i, t \geq ETA_i \quad (56)$$

The constraints (40) – (55) are relevant to staff assignment decision-

making, which characterizes the assumptions and requirements while forming the maintenance team, assigning maintenance technicians and arranging each individual's maintenance roster. Constraint (40) ensures that any maintenance task associated with aircraft i can be conducted as long as the aircraft is parked in the hangar. The situation in which the aircraft is leaving but the maintenance tasks of aircraft is scheduled to be conducted at the beginning of shift t , i.e. $y_{ist} = 1, p_{it} = 1, out_{it} = 1$, is not a possible scenario, since Constraint (54) prescribes that the roll out operation cannot be triggered before completing all maintenance tasks for the aircraft, and therefore Constraint (40) is not involved in the aircraft rolling out decision variable. Constraint (41) imposes the precedence relations between the maintenance tasks associated with aircraft i , implying that the subsequent maintenance tasks cannot be conducted before the preceding tasks have been finished. Constraints (42) – (44) ensure that each maintenance task has enough qualified maintenance technicians, and the number of capable technicians does not exceed the required number for a particular task to avoid wastage of manpower. Constraint (45) prescribes that no more maintenance technicians are assigned to the finished maintenance tasks. Constraint (46) ensures that the maintenance technicians cannot be allocated to aircraft not parked in the hangar. Constraints (47) – (48) determine if the maintenance task s associated with aircraft i have finished by shift t . Constraint (49) prescribes that the maintenance time of each maintenance task is equal or larger than the required maintenance time. Constraints (50) – (54) are the regulations in assigning multi-skill maintenance technicians. Constraints (50) and (51) ensure that a technician can be assigned to shift t if the particular individual is available, and the working time of the individual cannot exceed the prescribed working time limit of that division div . Constraint (52) prescribes that an individual maintenance technician cannot undertake maintenance tasks in two consecutive shifts to ensure the technician has sufficient rest. Constraint (53) ensures that the aircraft cannot leave the hangar before completing all the maintenance tasks associated with that aircraft. Constraints (54) – (56) prescribes the manpower assignment decision variables, maintenance task indication decision variables as binary.

3.4. Branching strategy

Considering the large number of binary variables involved in the mathematical model, difficulties in updating the incumbent solutions and lower bounds are expected during the optimization process, as eliminating unpromising solutions and duplicate/symmetric solutions is time-consuming. For example, an improper branching strategy may create a branching tree prescribing the manpower allocation variables on the top of the branching tree, which implies determining the manpower allocation before fixing the service period of the aircraft. Alternatively, an improper branching sequence may indicate the position-related variables at the top of the branching tree, which determines the position of the aircraft without confirming their respective service period. Such unwise or default branching strategies may result in adjustment of the relevant decision variables or pruning unpromising subtrees from the top of the branching tree in an inefficient way. Therefore, given the hierarchal structure of the binary decision variables, a branching strategy that caters to the features of mathematical model can be developed to avoid inefficient default branching strategies.

The hierarchal structure of the binary variables can be listed in a descending order as follows:

- 1) determine the service period (parking period) of each aircraft with the associated $in_{it}, out_{it}, p_{it}$ and out_{iT}^* . Normally the value of out_{iT}^* for all incoming maintenance demands equals zero, implying that the maintenance demands can be delivered by the end of the planning period;
- 2) After determining the service period of each incoming aircraft, the

non-overlapping constraints and movement path blocking constraints are imposed to validate the tentative service period. Therefore, the value of the binary variables h_{ijt} , L_{ij} , R_{ij} , U_{ij} and b_{ijkt} are branched to determine the coordinates (x_i, y_i) of each aircraft, and to examine the hangar capacity and clearance in the movement path in the tentative parking positions;

- 3) Upon determining the service period and geometric positions of each aircraft, the allocation of multi-skill maintenance technicians to each maintenance tasks is conducted. In the problem instances with overwhelming maintenance demands or in peak maintenance periods, the negative impact of poor coordination among the service time decisions, parking positions and manpower allocations is amplified, which shows in the interdependent relationships among these three core decision issues in the hangar maintenance planning problem.

For these instances with high demand, branching on the binary variables and updating bounds can be trapped for a long time, and therefore a branching strategy tailored for this problem is proposed to assist the branch-and-bound algorithm in searching for the incumbent solutions. The branching priorities assigned to the binary variables follow the hierarchical structure of the mathematical model, i.e. the priorities are assigned to service period-related, position-related and manpower-related binary variables in decreasing order. In this section, we conduct computational experiments to find the performance of the MILP model and the proposed branching strategy in solving small-sized instances.

4. A two-stage optimization approach

The MILP model presented in Section 3 involves a great number of geometric constraints, resource constraints and respective decision variables, which makes medium- and large-scale instances intractable by the default solver. In this section, two-stage optimization based on model decomposition is presented to provide a good quality solution for tackling the planning problem.

4.1. Decomposition of original model

In the original MILP model, the timing constraints, geometric constraints and resources constraints are integrated to reflect the interdependent relations among the three-core decision-making elements, namely the parking period of each aircraft in the hangar, the parking stand position and the assignment of technicians. As the mathematical model involves a larger number of binary decision variables, the branching process takes up a large amount of time. Updating bounds or finding new incumbent solutions become difficult for medium- to large-size instances, since the default branching strategies provided by the solver CPLEX are incapable of analyzing the complex relations and practical meaning behind the set of binary variables. Specifically, a hierarchical structure exists in the MILP model, and branching progress may be trapped into investigating an unpromising pending solution for a relatively long time. The hierarchical structure of the mathematical model can be presented as follows: the service period (parking period) of each aircraft is determined according to their ETA and ETD first. Afterwards, the non-overlapping constraints and movement path blocking constraints are imposed to validate the tentative service period, so as to determine whether the hangar has sufficient capacity for aircraft parking and clear movement paths to their dedicated positions. If an infeasible solution is returned under the tentative determined service period, then one or more aircraft's service period or parking positions need to be adjusted in order to align with the geometric constraints. Our previous research and computational analysis revealed that finding feasible parking plans for a single time or multi-periods is challenging while dealing with large numbers of incoming aircraft (Qin et al., 2019; Qin et al., 2018). In this extended model, we

incorporate the multi-skill technician assignment problem, which is another bottleneck in fulfilling hangar maintenance demand under the MRO mode. The hierarchical structure incorporating multi-skill technician assignment in the extended model makes the branching strategy incapable of tackling the medium-size integrated instance, especially when dealing with an instance with overwhelming maintenance demands and limited maintenance resources (hangar space and manpower). In this regard, we propose a two-stage optimization approach inspired by the decomposition method to reduce the complexity while optimizing the original model.

The original model is decomposed into two subproblems. Considering that solving a pure multi-period aircraft movement planning problem or a pure multi-skill staffing problem is tractable within a reasonable time by the methods in (Chen et al., 2017; Qin et al., 2019); we decompose the integrated problem into aircraft movement planning in the hangar and the multi-skill staffing problem, and set a time limit (1,800 s) as the stopping criteria for each subproblem in the iteration process. The first-stage problem consists of all the geometric constraints to determine the parking stands, movement path blocking and service time of each aircraft in the hangar. In the second-stage problem, the resource constraints related to the multi-skill technician assignment problem is included. To ensure the connectivity between two subproblems, an iterative process with linkage constraints is proposed to develop an integrated solution between geometric- and manpower-related decision making. A detailed description of the two-stage optimization approach is discussed in Section 4.2. The characteristics of the proposed approach is that the service time solution given from the first stage is flexible and can be adjusted during the optimization process of the second stage problem. The overview of the optimization procedures is presented in Algorithm 1.

4.2. First stage problem

The decision variables and constraints related to decision-making in the service planning period and the geometric aspects are included, i.e. to determine the parking period of an aircraft, parking position and movement path clearance. Instead of determining the working schedule of each maintenance task associated with the aircraft, the first stage problem only determines the service and parking time period for each aircraft. The first stage problem's optimization process consists of two scenarios:

- 1) The first scenario is the problem initialization. It is assumed that the manpower supply is sufficient to meet all the maintenance tasks at any time, which means that the maintenance tasks for each aircraft can be conducted consecutively during the whole planning period. After determining the parking periods, the parking stands and the movement paths for all aircraft, these decisions are passed to the second stage problem for the assignment of maintenance technicians, as well as feasibility checking;
- 2) The second scenario is the iteration process after problem initialization. After inputting the initial solution (or the solution in previous iterations), mismatch between the problems in the first and second stages might occur. In the problem initialization stage, it is assumed that the manpower supply is sufficient to meet all the maintenance tasks at any time along the planning period. However, the initial solution given in the first stage problem usually does not comply with the staff assignment problem in the second stage at the problem initialization step, which means that the manpower supply cannot meet all maintenance tasks with the predetermined desired time windows given by the first stage problem. Given that the available manpower is not able to meet the desired service period, the service time decision has to be adjusted to align with the manpower supply. A possible way is to extend the aircraft parking time so as to allow sufficient time to finish the maintenance tasks related to the aircraft. The alignment between the first stage and second

stage problem can be found after the adjustment of the parking time of the aircraft.

4.3. Second stage problem

The decision variables and constraints related to service time scheduling and multi-skill technician assignment form the second stage problem, which determines the number of technicians that serve the maintenance tasks within the tentative service period determined in the first stage problem. In the second stage problem, the fulfillment of maintenance tasks is specified, based on the tentative service period determined by the first stage problem. The optimization process of the second stage problem consists of two scenarios:

- 1) The second stage problem is able to identify a feasible solution with the service period determined in the first stage problem, which means that the manpower available is able to fulfill all the maintenance tasks within the desired service period given by the first stage problem. Service time adjustment is not required, and a feasible solution can be obtained.
- 2) Infeasibility returns after solving the second stage problem with the service period determined in the first stage problem. Under such circumstances, the service periods have to be adjusted in order to impose a longer time for the aircraft to stay in the hangar to fulfill its maintenance task.

The auxiliary binary decision variables are introduced in the second stage problem.

out_break_i	1, if the roll out time of aircraft i is amended from the outcome of the second stage problem, and 0 otherwise
$out *_break_i$	1, if fail to finish the maintenance tasks for aircraft i from the outcome of the second stage problem, and 0 otherwise

The service time decisions for all incoming aircraft made in the first stage problem serve as the initial solution for the second stage problem. The initial solution consists of $\{in_{it}, out_{it}, out_{iT}^*\}$ for all aircraft, imposed as “soft constraints” in the second stage problem, which means that the tentative service time decision determined in the first stage problem can be adjusted if deemed necessary when the manpower is insufficient to serve all the maintenance tasks within the tentative service period determined in the first stage problem. The relevant constraints are as follows:

$$in_sp_{it} \geq \overline{in_{it}} - in_break_i, \forall i \in A_T, \forall t \geq 0 \quad (57)$$

$$out_sp_{it} \geq \overline{out_{it}} - out_break_i, \forall i \in D_t, \forall t \geq 0 \quad (58)$$

$$out_{T^*}sp_{it} \geq \overline{out_{iT}^*}, \forall i \in D_t, \forall t \geq 0 \quad (59)$$

Constraints (57)–(59) prescribe the tentative service time determined by the first stage solution, which are changeable if deemed necessary, as mentioned above. The adjustable service time decisions are the roll in and roll out times of aircraft, which allows extensions of the staying time of aircraft in the hangar so as to have more time to finish the maintenance tasks.

4.4. Linkage constraints for the two-stage problem

$$\left(\sum_{t \geq ET_{D_i}} out_{it} \cdot t - \sum_{t \geq ET_{A_i}} in_{it} \cdot t \right) + M \cdot \left(1 - \sum_{t \geq ET_{A_i}} in_{it} \right) + M \cdot \left(1 - \sum_{t \geq ET_{D_i}} out_{it} \right) \geq MTime_i^{REVISED}, \forall i \in A_T$$

The linkage constraint (60) is added into the first stage problem and

resolved again in the next iteration. The revised maintenance time $MTime_i^{REVISED}$ can be derived by the revised roll out time of aircraft i out_sp_{it} from the second stage problem and the original roll in time $\bar{in}_{it}$ from the first stage problem. $MTime_i^{REVISED} = out_sp_{it} - \bar{in}_{it}$ for the aircraft needs extension of the staying time in the hangar. Another enhanced strategy of service time window adjustment is postponement of the roll in time of the aircraft that cannot be serviced under manpower supply shortage situations, with flexible adjustment of maintenance time. The adjusted roll in time can be determined in the second stage problem by relaxing the original roll in time in_{it} constraint derived from the first stage problem. Specifically, the second stage problem evaluates the potential improvement of postponing the roll in time, together with the necessary maintenance time $MTime_i$ adjustment, maintenance demand fulfillment tardiness consideration. In this regard, the postponement of the original roll in time expects: 1) roll in time postponement only; and 2) roll in time postponement together with the extension of maintenance time $MTime_i$. Postponing of roll in time may result in on-time or delay in fulfilling the maintenance demands, subject to corresponding Estimated Time of Departure (ETD) and available manpower supply. Selection of two strategies is determined by evaluating its contribution to the objective function value, i.e. the relevant tardiness penalty cost/failure of maintenance fulfillment and manpower cost after time window adjustment. The second stage problem determines the adoption of adjustment strategies, by calculating their respective contribution to the objective value. The revised service time windows associated with roll in time, roll out time and maintenance time decisions are passed to the first stage problem for feasibility checking. After adding the revised maintenance time constraint and revised roll in time (when appropriate) back to the first stage problem, two possible outcomes for the service time decisions are expected: 1) a revised service time period has been found (a revised roll in & roll out time is determined), and the aircraft can be serviced during the planning period; 2) after imposing the revised staying time constraint, the aircraft is rejected for maintenance service (the aircraft is not rolled into the hangar), due to the increase of delay cost induced by the extended staying time or blocking with other aircraft.

Algorithm 1. Two-stage optimization approach for the hangar maintenance planning problem

Notations	Meanings
in_{it}^1, out_{it}^1	Roll in and roll out time for aircraft determined in first stage problem
in_{it}^2, out_{it}^2	Roll in and roll out time for aircraft determined in second stage problem
A_T	All maintenance aircraft during the planning period
in_break_i	Indicator of adjusting roll in time of aircraft i from the outcome of the second stage problem
out_break_i	Indicator of adjusting roll out time of aircraft i from the outcome of the second stage problem
$RollOutT^*_i$	Indicator of failing to deliver the aircraft i from the outcome of the second stage problem
$MTime_i^{REVISED}$	The revised staying time requirement of aircraft i after optimizing the second stage problem
1:	Solve the first stage problem (with the revised movement times derived from the second stage problem, if applicable) and obtain the first stage service time decision, including the service time and roll in & roll out movement time decisions. If the revised roll in time derived by strategy 2 induces infeasibility, return to the original roll in time with $MTime$ extension derived by strategy 1.
2:	Input the first stage decision into the second stage problem. Solve the second stage problem to determine the staff rosters along the planning period.
3:	If the second stage solution is infeasible, the misalignment between service time arrangement and manpower supply exists.
4:	Evaluate the potential improvement of objective value of adopting two service time window adjustment strategies.
5:	Determine one adjustment strategy for each aircraft that cannot be serviced due to manpower shortage
6:	For i in A_T


```

7:      If  $out\_break_i = 1$  &  $in\_break_i = 0$ 
      (Adoption of Service adjustment strategy 1: The roll out time of
      aircraft  $i$  is extended, and the roll in time remains unchanged)
8:      Calculate the revised staying time required for the aircraft
 $i.MTime_i^{REVISED} = \sum_{t \geq ETD_i} out_{it}^2 \cdot t - \sum_{t \geq ETA_i} in_{it}^1 \cdot t$ 
9:      If  $i.MTime_i^{REVISED} > \sum_{t \geq ETD_i} out_{it}^1 \cdot t - \sum_{t \geq ETA_i} in_{it}^1 \cdot t$ 
10:      Generate the revised staying time constraint for the
      aircraft  $i$ . Add the maintenacne time constraint to the first stage
      problem
11:      If  $in\_break_i = 1$ 
      (Adoption of Service adjustment strategy 2: The roll in
      time is postponed)
12:      If  $out\_break_i = 1$ 
      (Roll out time is extended)
13:      Calculate the revised staying time required for the aircraft
       $i$ 
14:       $MTime_i^{REVISED} = \sum_{t \geq ETD_i} out_{it}^2 \cdot t - \sum_{t \geq ETA_i} in_{it}^2 \cdot t$ 
15:      If  $MTime_i^{REVISED} > \sum_{t \geq ETD_i} out_{it}^1 \cdot t - \sum_{t \geq ETA_i} in_{it}^1 \cdot t$ 
16:      Generate the revised staying time constraint for the aircraft  $i$ .
      Add the constraint and the revised roll in time to the first stage
      problem
17:      If  $out\_break_i = 0$ 
      (Roll out time remains unchanged)
18:      Calculate the revised staying time required for the aircraft
       $i$ 
19:       $MTime_i^{REVISED} = \sum_{t \geq ETD_i} out_{it}^1 \cdot t - \sum_{t \geq ETA_i} in_{it}^2 \cdot t$ 
20:      If  $MTime_i^{REVISED} > \sum_{t \geq ETD_i} out_{it}^1 \cdot t - \sum_{t \geq ETA_i} in_{it}^2 \cdot t$ 
21:      Generate the revised staying time constraint for the
      aircraft  $i$ . Add the constraint and the revised roll in time to the first
      stage problem
22:      If  $RollOut_i * r_i^2 = 1$ 
      (The maintenance demands of aircraft  $i$  cannot be fulfilled
      till the end of planning period)
23:      Generate the  $RollOut_i^* \geq 1 - M \cdot \left(1 - \sum_{t \geq ETA_i} in_{it}\right)$  constraint.
      Add the constraint to the first stage problem
24:      Go to Step 1.
25:      Else if the second stage solution is feasible
26:      The current service time solution can find feasible technician
      assignment plan. Go to Step 14.
27:      End

```

5. Computational experiments

In this section, we describe the ways of generating problem instances based on real data collected from an aircraft maintenance company, and analyse the numerical experiment results. The approaches presented in the methodology section were programmed in C# and implemented in Visual Studio 2010, using a computer with an Intel Core i7 processor, at 3.6 GHz with 32 Gb of RAM. The MILP model was solved by the optimizer CPLEX 12.7 serial model.

5.1. Description of test instances

The problem instances were generated from the maintenance demand information collected from an independent aircraft heavy maintenance company located in Hong Kong, serving over 50 clients, including airlines, business jet companies and utility aircraft companies. The necessary information on the maintenance demands, including the estimated arrival time (ETA), estimated time of departure (ETD) also known as the desirable service completion time, aircraft type and maintenance checks type of each maintenance demand, was collected from multiple clients over 157 days to create instances. We utilized these sets of data to generate problem instances in our previous study (Qin et al., 2019; Qin et al., 2017). The aircraft to be maintained are classified into three categories according to their physical size, as presented in Table 3.

We refer to Ertogral and Öztürk (Ertogral and Öztürk, 2019)'s principle while determining the cost of maintenance technicians, which prescribes the cost on an hourly rate basis. Considering the multi-skill maintenance technician setup in this problem, it is reasonable to assume that an individual technician equipped with more maintenance skills has a higher wage rate, regardless of the maintenance tasks assigned to the technician in actual implementation. Such cost setting encourages assigning senior maintenance technicians, i.e. the individuals equipped with more maintenance skills and senior maintenance licenses, to the maintenance tasks requiring senior technicians to avoid improper utilization of manpower and wastage of manpower. The number of maintenance technicians required for the different tasks in different categories are listed in Table 4. To examine the performance of the proposed two-stage optimization approach, the following parameters are adjusted across the problem instances: 1) the skill levels of the maintenance technicians; 2) number of available staff; 3) required maintenance service and associated maintenance tasks and 4) shift settings.

For the three penalty costs, chance loss cost (Penalty Cost 1: Penalty for not serving aircraft i during planning period (per request)), tardiness cost (Penalty Cost 2), as well as the failure to deliver aircraft with all maintenance tasks completed upon the end of planning period (Penalty Cost 3), each maintenance demand has its own set of costs according to the maintenance type, aircraft type and the required maintenance skill level. In detail, the chance loss cost (Penalty Cost 1) refers to the situation in which the maintenance company does not have enough maintenance capacity to serve the maintenance demand and rejects the demand. Originally, the profit in completing the maintenance service for an aircraft is prescribed as two times of the manpower cost of conducting the task, and the chance loss cost of rejecting the service request is prescribed as the profit of completing the maintenance service for the aircraft. The tardiness cost is calculated on a time (minute) basis, which is induced whenever maintenance service for the aircraft is finished after the desirable delivery time (ETD). The cost of failure to deliver the aircraft by the end of planning period is prescribed as a portion of chance loss cost (Penalty 1) in the computational experiments.

5.2. Computational results

Two sets of computational experiment results are presented in this section. Section 5.2.1 reports the computational results of the basic MILP model in small- size instances. Afterwards, the two-stage optimization approach was examined by the medium- and large-size instances, and the results of numerical experiment are reported in Section 5.2.2.

5.2.1. Model and branching strategy's evaluation

In this section, the performance of the proposed MILP formulation and branching strategy presented in Section 3.4 is examined. The performance comparison between the original model and the model with the branching strategy is shown in Table 5. The number of shifts for a single day is prescribed as three, and the length of each shift is 8 h, which aligns with the normal staff rostering setting. The first column in the table denotes the name of the instance. The maintenance demand data collected from the maintenance company are organized to create

Table 3
Aircraft Size Classification.

Classification	Total Number	Aircraft models
Small-sized	10	G200 CL600 CL605 F900LX F2000EX F2000LX ERJ135 F7X G450 GIV
Medium-sized	11	GL5T G550 G5000 G6000 G650 A318 ERJ190 A319 A320 B738 A321
Large-sized	2	A332 A333

Table 4
Generic number of required maintenance technicians.

	Small Aircraft	Medium Aircraft	Large Aircraft
Minor Maintenance Task	2	3	5
Medium Maintenance Task	3	4	6
Major Maintenance Task	4	6	9

different groups of problem instances, which are divided into subsections to create problem instances of different sizes or with different parameters. The maintenance demand data are sorted on a monthly basis, and further divided into multiple sets of problem instances within the month. The instance's name is presented in the form of "number of requests_maintenance tasks's demand level (number of planning days)", e.g. 5_1(14) indicates that such instance includes 5 incoming aircraft to be serviced with the standard manpower demand, which covers 14 days. To investigate the performance of the solution approach, instances covering half a month to two months are created to examine the impact of the planning period length and number of maintenance demands. The original model and the model incorporating the branching strategy mentioned in the beginning of this section solve the problem instances. The second column shows the number of binary variables involved in the mathematical model, and the third column denotes the preprocessing time before implementing the branch-and-bound algorithm embedded in CPLEX. To examine the impact of increasing the manpower requirement associated with the maintenance task (the number of qualified technicians required for maintenance tasks) on the objective value and the computational performance, an adjustment on number of maintenance technicians required for conducting maintenance task is implemented to create a variation of the manpower demand. Specifically, Instance x_1(planning days) denotes the original problem instance with the maintenance team size aligned with Table 4, and Instance x_2(planning days) and Instance x_3(planning days) refers to the problem instances with same maintenance tasks setting on each aircraft as in Instance x_1(planning days), but each maintenance task in Instances x_2(planning days) and x_3(planning days) requires more manpower inputs in a progressive manner, i.e. Instance x_3(planning days) prescribes that more maintenance technicians are required to conduct the identical tasks than in Instance x_2(planning days). Therefore, the same group of instances with manpower requirement variation have the same number of binary variables. The preprocessing time refers to the time spent on initializing the model formulation in the optimizer, and the stopping criteria for solving each problem instance in both models are prescribed with a time limit 3,600 s.

Table 5 shows that both models cannot solve all problem instances optimally within one hour. Nevertheless, a minor advantage of the branching strategy is shown, compared with the original model in most instances, in terms of the best-known solution and the optimality gap

for some instances. In particular, the branching strategy finds better incumbent solutions or tightens the optimality gap over the original model in the problem instance with a moderate number of maintenance demands or higher manpower requirement, e.g. 8_3(22) and 10_2(32), while such advantages do not appear in solving the small-sized, challenging instances with a large number of maintenance demands and high manpower requirements, e.g. 5_1(14), 5_2(14), 5_3(14) and 15_3(63). Therefore, the acceleration effects of the branching strategy only emerge when the maintenance demand grows to a moderate level. In order to identify the optimal solutions, we further extend the time limit to 2 h (7,200 s) and 5 h (18,000 s). The relevant results are presented in Table 6 and Table 7 respectively. It is noted that the optimality gaps of most instances are tightened and Instance 5_2(14) was solved to optimal by around 3 h, while for large-scale instances the optimality gap remains large ones in both 2 h and 5 h limits settings. Extensions of the time limits to 10 h (36,000 s) for the small- and medium-sized instances were carried out afterwards to investigate the improvement of incumbent solutions. The results reported in Table 8 show that there is no additional instance solved to optimal while extending time limit to 10 h. The gap reductions for most instances are minimal (except a single instance 8_22(22) solved by MILP without branching strategy). In this regard, extending the computational time to a relatively long period did not help the branch-and-bound algorithm in achieving a moderate improvement in finding optimal solution or in narrowing the optimality gap within reasonable time, which implies that the branch-and-bound algorithm is trapped in the searching process while the problem size grows with large number of binary variable involved and the incapability of the updating bounds. It is noted that great number of binary decision variables are created in each instance, and the number of binary variables grows significantly along with extending the planning period as well as increasing in the number of maintenance tasks. Besides the number of maintenance requests in the instances, the difficulties of tackling instances also lie in the distribution of arrivals, maintenance tasks and precedent relations associated with the aircraft, as well as the manpower input requirements of the maintenance tasks. The analysis on the same group of problem instances with a variation of manpower requirements on the maintenance task reveals an increasing trend in the objective value, which is associated with the increase of the manpower cost and rejection of the maintenance requests due to the insufficient manpower supply. The rejection of maintenance requests is also reflected in the shrinkages of the optimality gap, as the branching efforts are saved for examining the service period, parking stand allocation, geometric relations with other aircraft and the manpower assignment on the rejected maintenance requests. The minor advantages of the branching strategy demonstrate the hierarchical structure of the mathematical model that imposes computational difficulties in tackling the problem instances. However, the results in solving instances also reflect the inefficiency of tackling

Table 5
Comparison between original model and model with branching strategy (1-hour time limit).

Instance	Binary Variables	Preprocessing Time (s)	MILP Model without Branching Strategy				MILP Model with Branching Strategy			
			Upper Bound	Lower Bound	Gap (%)	CPU (s)	Upper Bound	Lower Bound	Gap	CPU (s)
5_1(14)	11,734	0.48	43719.00	38198.50	12.63	3,600	41583.00	35235.32	15.22	3,600
5_2(14)		0.40	58266.00	45532.41	21.85	3,600	57930.00	43953.50	24.13	3,600
5_3(14)		0.42	54234.00	54234.00	0	103.82	54234.00	50490.13	6.90	3,600
8_1(22)	67,927	2.52	83595.00	12357.97	85.21	3,600	83595.00	12343.76	85.23	3,600
8_2(22)		2.76	95115.00	40431.77	57.49	3,600	95091.00	40421.21	57.49	3,600
8_3(22)		2.69	113091.00	71447.20	36.82	3,600	112827.00	111968.31	0.76	3,600
10_1(32)	116,022	4.44	129351.75	11640.78	91	3,600	82311.75	11631.62	85.87	3,600
10_2(32)		4.41	176055.75	35495.72	79.84	3,600	38577.75	35503.16	7.97	3,600
10_3(32)		4.60	194877.75	57955.29	70.26	3,600	166002.75	57979.18	65.07	3,600
15_1(63)	162,657	8.29	195156.00	59608.39	69.46	3,600	216510.00	59600.51	72.47	3,600
15_2(63)		8.61	413802.00	87353.60	78.89	3,600	226206.00	87679.71	61.24	3,600
15_3(63)		7.96	490815.00	132279.25	73.05	3,600	490815.00	132279.25	73.05	3,600

Table 6

Comparison between original model and model with branching strategy (2-hour time limit).

Instance	MILP Model without Branching Strategy					MILP Model with Branching Strategy				
	Upper Bound	Lower Bound	Gap (%)	Gap Reduction (%)	CPU (s)	Upper Bound	Lower Bound	Gap	Gap Reduction (%)	CPU (s)
5_1(14)	43815.00	38295.63	12.60	0.24	7,200	41583.00	35301.04	15.11	0.72	7,200
5_2(14)	52626.00	46485.90	11.67	46.59	7,200	57762.00	44087.45	23.67	1.91	7,200
5_3(14)	54234.00	54234.00	0	N/A	112.17	54234.00	50549.72	6.79	1.59	7,200
8_1(22)	51672.00	12349.34	76.10	10.69	7,200	15069.00	12355.05	18.01	78.87	7,200
8_2(22)	43173.00	40433.41	6.35	88.95	7,200	94947.00	40652.61	57.18	0.54	7,200
8_3(22)	112995.00	71462.31	36.76	0.16	7,200	112755.00	112017.30	0.65	14.47	7,200
10_1(32)	114927.75	11647.43	89.87	1.24	7,200	82311.75	11631.62	85.87	0.00	7,200
10_2(32)	99543.75	35523.89	62.74	21.42	7,200	38289.75	35510.33	7.26	8.91	7,200
10_3(32)	164178.75	57994.58	64.68	7.94	7,200	115815.75	58048.05	49.88	23.34	7,200
15_1(63)	194943.00	59608.38	69.42	0.06	7,200	215766.00	59600.51	72.38	0.12	7,200
15_2(63)	180966.00	88169.59	51.28	35.00	7,200	204387.00	88169.59	56.86	7.15	7,200
15_3(63)	490815.00	132279.25	73.05	0.00	7,200	490815.00	132279.25	73.05	0.00	7,200

Table 7

Comparison between original model and model with branching strategy (5-hour time limit).

Instance	MILP Model without Branching Strategy					MILP Model with Branching Strategy				
	Upper Bound	Lower Bound	Gap (%)	Gap Reduction (%)	CPU (s)	Upper Bound	Lower Bound	Gap	Gap Reduction (%)	CPU (s)
5_1(14)	43455.00	38380.12	11.68	7.30	18,000	41223.00	35399.63	14.13	6.49	18,000
5_2(14)	50658.00	50658.00	0	100.00	9938.45	52983.00	45587.12	13.96	41.02	18,000
5_3(14)	54234.00	54234.00	0	N/A	117.34	54234.00	50583.23	6.73	0.88	18,000
8_1(22)	50928.00	12576.47	75.31	1.04	18,000	14757.00	12596.01	14.64	18.71	18,000
8_2(22)	41901.00	40839.79	2.53	60.16	18,000	94947.00	40704.27	57.13	0.09	18,000
8_3(22)	112707.00	112227.00	0.43	98.83	18,000	112755.00	112059.00	0.62	4.62	18,000
10_1(32)	107748.00	11652.92	89.19	0.76	18,000	82311.75	11667.54	85.83	5.68	18,000
10_2(32)	41169.75	35524.41	13.71	78.15	18,000	38025.75	35523.82	6.58	9.37	18,000
10_3(32)	125778.75	58071.32	53.83	16.77	18,000	103020.75	58011.23	43.69	12.41	18,000
15_1(63)	161358.00	59663.39	63.02	9.22	18,000	195582.00	59724.72	69.46	4.03	18,000
15_2(63)	180531.00	88395.37	51.04	0.47	18,000	179598.00	88172.66	50.91	10.46	18,000
15_3(63)	274785.00	137813.66	49.85	31.76	18,000	266220.00	137814.45	48.23	33.98	18,000

Table 8

Comparison between original model and model with branching strategy (10-hour time limit).

Instance	MILP Model without Branching Strategy					MILP Model with Branching Strategy				
	Upper Bound	Lower Bound	Gap (%)	Gap Reduction(%)	CPU (s)	Upper Bound	Lower Bound	Gap	Gap Reduction(%)	CPU (s)
5_1(14)	43455.00	38476.00	11.46	1.88	36,000	41223.00	35495.28	13.89	1.70	36,000
5_2(14)	50658.00	50658.00	0	N/A	10091.90	52911.00	45682.02	13.66	2.15	36,000
5_3(14)	54234.00	54234.00	0	N/A	148.26	54234.00	50664.22	6.58	2.23	36,000
8_1(22)	50592.00	12576.72	75.14	0.23	36,000	14757.00	12633.16	14.39	1.71	36,000
8_2(22)	41541.00	41178.60	0.87	65.61	36,000	94947.00	40889.02	56.93	0.35	36,000
8_3(22)	112683.00	112275.00	0.36	16.28	36,000	112731.00	112059.00	0.6	3.23	36,000

the problem solely with the branching strategy. It can be inferred that extending the length of the planning period also influences the complexity of solving a single instance as it determines the scale of the time-related decision variables.

5.2.2. Two-stage optimization approach evaluation and the variations of manpower requirements

The computational results in Section 5.2.1 have demonstrated the minor advantages of the branching strategy in tackling the instances over the original model. However, it is intractable to deal with the instances solely by the MILP model with the branching strategy as the instances cannot be solved optimally after meeting the stopping criterion. In this section, we further implement the computational experiment to examine the performance of the two-stage optimization approach presented in Section 4. In the report of the computational results, the performance of the MILP model incorporating the branching strategy is compared with the two-stage optimization approach. We examine the performance of the two-stage optimization approach with

a difference in the shift setting in one day, i.e. setting 3 shifts in one day and 4 shifts in one day. The MILP model tailored for the two-stage optimization approach is embedded in solving the first stage problem and the second stage problem with the branch-and-bound algorithm. The time limit for solving each first and second stage problem is 1,800 s.

Table 9 shows the computational results of the two-stage optimization approach. The seventh to eighth columns of Table 9 show the binary variables involved in the problems in two stages, and the number of iterations, respectively. The CPU and objective values are used to compare the performances between the model with the branching strategy and the proposed two-stage approach. The two-stage optimization approach manages to obtain solutions with good quality within around an hour, compared with the MILP model with the branching strategy. The advantages of two-stage approach are reflected in the objective values in tackling problem instances across different instance sizes. It is found that setting 4 shifts per day does not reduce the manpower cost. The required computational time and model scale

Table 9
Computational results in solving instances by two-stage optimization approach.

Instance	Shift Setting	MILP Model with Branching Strategy				Two-stage Optimization Approach					
		Upper Bound	Lower Bound	Gap	CPU	Binary Variables in Stage 1 Problem	Binary Variables in Stage 2 Problem	Iteration	Generated Constraints	Objective Value	CPU
5_1(14)	3	41583.00	35235.32	15.22	3600	5487	6721	1	4	41391.00	2110.34
5_2(14)		57930.00	43953.50	24.13	3600			1	10	52650.00	1963.34
5_3(14)		54234.00	50490.13	6.90	3600			1	7	54234.00	468.64
8_1(22)		83595.00	12343.76	85.23	3600	34,100	35,347	1	1	15333.00	3522.25
8_2(22)		95091.00	40421.21	57.49	3600			1	3	42477.00	3648.89
8_3(22)		112827.00	111968.31	0.76	3600			1	6	112799.00	3138.09
10_1(32)		82311.75	11631.62	85.87	3600	68,320	49,953	1	1	157273.75	706.71
10_2(32)		38577.75	35503.16	7.97	3600			1	3	37633.75	3701.88
10_3(32)		166002.75	57979.18	65.07	3600			1	8	116343.75	3669.91
15_1(63)		216510.00	59600.51	72.47	3600	112,153	55,069	1	12	112857.01	4055.01
15_2(63)		226206.00	87679.71	61.24	3600			3	39	200205.00	2328.78
15_3(63)		490815.00	132279.25	73.05	3600			2	29	237273.00	4287.66
5_1(14)	4	51669.00	39097.39	24.33	3600	7372	9045	1	7	46431.00	2027.39
5_2(14)		69900.00	47597.70	31.91	3600			1	7	55863.00	2513.63
5_3(14)		58671.00	51300.52	12.56	3600			2	10	58561.00	701.28
8_1(22)		121539.00	38119.44	68.64	3600	45,505	35,310	3	20	101031.00	1909.08
8_2(22)		184776.00	57023.04	69.14	3600			3	20	124596.00	2057.15
8_3(22)		207024.00	62777.81	69.68	3600			2	16	147003.00	1890.51
10_1(32)		118938.75	34805.94	70.74	3600	91,700	43,629	3	27	86235.00	2519.36
10_2(32)		204966.75	55514.02	72.92	3600			2	22	141235.50	1951.91
10_3(32)		335211.00	65667.83	80.41	3600			3	25	147949.50	2338.98
15_1(63)		341013.00	59779.58	82.47	3600	149,634	73,565	5	43	150777.00	3726.58
15_2(63)		448527.00	85264.98	80.99	3600			2	34	221436.00	3327.33
15_3(63)		510036.00	130365.20	74.44	3600			1	20	289746.00	2940.70

for solving the same instances increases in the 4-shift per day setting. The strength of the two-stage optimization approach over the MILP model for large-scale or high demand instances implies the incapability of the MILP model in connecting the three independent core elements of decision-making, as the branch-and-bound algorithm is likely to probe the subtree associated with the infeasible solution repeatedly before updating the bounds, and the default branch-and-bound algorithm cannot infer the pattern of the unpromising solution from the previous pruned subtrees. The computational results of the MILP model with a branching strategy in 4-shift settings echoes the discussion and analysis in Section 5.2.1 that the branching strategy does not manifest advantages in solving small-sized problem instances, given the large optimality gaps recorded.

The effectiveness of the two-stage optimization approach reflects that service time window adjustment mechanisms, i.e. the staying time extension strategy and the roll in/roll out time postponement strategies on each incoming aircraft, successfully acts as an efficient connecting bridge between the two stage problem, as the main effect of insufficient manpower supply directly results in the tardiness in fulfilling the

maintenance tasks and an extension of the service time window. For the maintenance service company, facing hard instances for decision-making within a short period of time is common for real-world operations. Considering the tradeoff between the solution quality and efficiency, the proposed algorithm is capable of solving practical instances covering a relatively long planning period or high maintenance demands for practical use. The two-stage optimization approach can be considered as a reliable heuristic when the service providers are in need of better-quality solutions in less time than that provided in exact method by the commercial solver when the allowable computational time is limited.

5.2.3. Enhancement of maintenance technician's skill

The computational experiment described in Sections 5.2.1 and 5.2.2 focus on the computational efficiency in dealing with the problem instances without considering the variation of manpower supply in this section. We conduct an analysis to discover the impact of the variation of the manpower supply on the solution, which provides better understanding for the service provider to better exploit the maintenance capability and realize a profitable portfolio.

Table 10
Enhancement of maintenance manpower supply.

Instance	Before Manpower Supply Enhancements				Manpower Supply Enhancement 1				Manpower Supply Enhancement 2			
	Iteration	Generated Constraints	Objective Value	CPU	Iteration	Generated Constraints	Objective Value	CPU	Iteration	Generated Constraints	Objective Value	CPU
5_1(14)	1	4	41391.00	2110.34	1	2	17349.00	2231.89	1	1	14208.00	3018.50
5_2(14)	1	10	52650.00	1963.34	1	4	37860.00	596.00	1	2	24612.00	1858.71
5_3(14)	1	7	54234.00	468.64	1	7	52770.00	1936.16	1	2	35340.00	3620.83
8_1(22)	1	1	15333.00	3522.25	0	0	13341.00	649.23	0	0	12717.00	80.21
8_2(22)	1	3	42477.00	3648.89	0	0	36237.00	29.66	1	1	34269.00	3655.21
8_3(22)	1	6	112799.00	3138.09	1	1	53133.00	3658.81	0	0	52509.00	1839.39
10_1(32)	1	1	157273.75	706.71	0	0	13041.75	36.26	0	0	12561.75	36.22
10_2(32)	1	3	37633.75	3701.88	0	0	36049.75	439.25	1	1	35697.75	3652.43
10_3(32)	1	8	116343.75	3669.91	1	1	59169.75	3651.27	0	0	58065.75	1821.71
15_1(63)	1	12	112857.01	4055.01	1	1	63105.00	4082.78	0	0	61545.00	2003.39
15_2(63)	3	39	200205.00	2328.78	1	7	123720.00	3873.09	3	10	96675.00	7871.68
15_3(63)	2	29	237273.00	4287.66	5	18	181977.00	2710.78	0	0	121017.00	1896.01

Two settings of manpower supply enhancement are employed. In Table 8, the column “Manpower supply enhancement 1” refers to the setting of “minor” manpower supply enhancement, and the “Manpower supply enhancement 2” refers to the “major” manpower supply enhancement, respectively. The specification of maintenance demands, including the estimated arrival time, desirable deliver time, requirement of maintenance tasks (required skills and team size), remains unchanged through the manpower supply enhancement. Table 10 shows the computational results in enhancing the manpower supply for the instances. After enhancing the manpower supply progressively, it is found that the objective values of all instances have a decreasing trend when increasing the manpower supply across all skill levels. The iterations and generated additional constraints during the optimization progress also maintain a similar trend, which reflects that the difficulties in finding solutions have been reduced. The computational times for the larger size instances are reduced significantly compared with the original manpower supply. Nevertheless, the enhancement of manpower supply does not necessarily alleviate computational burden and shorten CPU times. It was found that solving an instance with particular manpower supply settings resulted in a high computational time, compared with the other manpower setting under the same instance. In this regard, we take instances 15_2(63) to further understand the cause of distinct high CPU time. Firstly, the instance setting of 15_2(63), including the maintenance tasks, manpower requirements and the available technicians, are investigated so as to interpret the impact of instance setting on computational efficiency as well the difficulties in resolving instances with different manpower supply settings.

We first analyse the impact of manpower enhancements on the model complexity. The number of available maintenance technicians in “Before Manpower Enhancements”, “Manpower Supply Enhancement 1” and “Manpower Supply Enhancement 2” settings are 12, 20, 32, respectively. For the mathematical model, the binary decision variable that directly associates with technicians assignment is $z_{mt,ist}$, together with maintenance task indication variable y_{ist} and auxiliary binary variables f_{ist} . As the available manpower supply in the problem instance increases, the above binary variables grow significantly, which is indicated in Table 11. Specifically, the number of binary variables involved in the second stage problem of 15_2(63) under three different settings (i.e. “Before Manpower Supply Enhancements”, “Manpower Supply Enhancement 1” and “Manpower Supply Enhancement 2”) are 55069, 84,141 and 127749, respectively.

The longest computational time falls on the setting “Manpower Enhancement 2”, which runs 3 iterations before meeting the stopping criterion. It was found that the CPU times of the last iteration in solving the second stage problem under 3 different manpower supply settings were 5.41, 1,800, 1,800 s respectively, which demonstrates an increases of model complexity and difficulties in updating the incumbent solutions after the enhancements of manpower supply. Meanwhile, the impact of manpower supply on the objective function has demonstrated in setting “Enhancement 2”, as a decreasing trend of objective values was recorded by increasing the manpower supply in Enhancement 1 & 2.

Under the “Manpower Enhancement 2” setting, the computational time of the second stage problem under the last iteration was 1,800, which results in one of the longest computational times among the 3 manpower supply settings. Although enhancements of manpower supply reduce the objective value (the portion related to tardiness costs), the alleviation of the computational burden is minimal. After comparing the manpower demands between 15_1 (63) and 15_2(63), the maintenance tasks in 15_2(63) require a greater number of eligible technicians to form a maintenance group than that of 15_1(63).

In addition, it was found that the computational time for solving 15_3(63) was lower than 15_2(63). Though 15_3(63) requires more manpower input in completing the same task, the solution space of the first stage problem in 15_3(63) is eliminated efficiently at the preliminary stage of optimization progress, since the respective strong constraints are generated in view of big difference between manpower

Table 11
Analysis on instance 15_2(63) under different manpower supply settings.

Manpower Settings	Iterations	Second Stage			
		First Stage	Second Stage		
		First Stage Objective (Costs of Tardiness and Demand Rejection)	CPU of the Last Iteration CPU	Second Stage Objective (Manpower Cost)	CPU of the Last Iteration
					Binary Variables
Before Manpower Supply Enhancements	1	189756.00	237.97	10449.00	55,069
Manpower Supply Enhancement 1	1	72984.00	50.13	50736.00	84,141
Manpower Supply Enhancement 2	3	21459.00	169.80	75216.00	127,749

supply and demands. For 15_2 (63), the manpower supply is enhanced but the solution space in the first stage problem cannot be easily tightened. Under the situation that the number of required manpower falls in medium range under the large-scale instance (15_x (63) series), the complexity in forming maintenance technicians' group grows significantly. The narrow difference between manpower supply and demands in 15_2 (63) weaken the efficiencies of the solution space tightening progress in the first stage problem.

While comparing the manpower cost, similar levels of manpower cost were recorded among the 15_x (63) series instances group, which indicates that the manpower was highly utilized under different settings. Based on the above-mentioned observations and analysis on the computational results, we can conclude that the manpower demands and the parameters related to available maintenance technicians supply become two significant factors that affect the model complexity and computational efficiency of two-stage optimization approach. Specifically, the medium range of manpower demands and high levels of manpower supply setting creates the most challenging instances, as the solution space tightening progress of the first stage problem becomes inefficient. The narrow difference between manpower demand and supply renders a large group of solutions around the manpower limit threshold. These solutions around the manpower limit threshold are eliminated after feasibility checking and matching between first and second stage problems, which creates significant computational burdens in the optimization progress.

6. Conclusions

Efficient service implementation in fulfilling increasing aircraft hangar maintenance demands has emerged as a crucial factor in an aircraft hangar maintenance service company. An aircraft hangar maintenance planning problem is studied under the MRO outsourcing mode as a new problem of aircraft maintenance planning. To fulfill the incoming maintenance demands from multiple airlines and other clients, maintenance planners need to optimize the service period, multi-period hangar parking plan and the manpower assignment roster together. The research work presented in this paper makes a novel contribution in closing the research gap in the maintenance optimization problem in the aviation industry. A mixed integer linear programming (MILP) model formulating the geometric constraints and manpower assignment constraints is proposed, so as to integrate and characterize the interdependent relations of decision-making. For the maintenance company, developing the hangar parking plan and multi-skill maintenance technicians' roster are the bottlenecks in fulfilling the maintenance demands and in implementing the maintenance tasks. The limited hangar capacity, flexible parking arrangement, and movement blocking, together with multi-skill maintenance technician rostering make the problem intractable by the MILP model with the default branching and bound method, as the scale of the model grows significantly. To tackle medium-to large-size problems, a two-stage optimization approach is developed to decompose the original model into two subproblems linked by constraints, and its effectiveness is examined afterwards. The developed two-stage optimization approach is examined by testing problem instances generated based on the maintenance demands data collected from industry. It is demonstrated that the proposed approach is able to provide a good quality solution within the time limits in tackling medium- to large-size instances. Afterwards, an analysis to study the effect and impact of the variation of the parameters in the maintenance demand and manpower supply is carried out. Given the interrelations among the three core elements of decision-making, the service capacity fluctuates with the incoming maintenance demands. The challenges in coordinating the maintenance resources are shown in the numerical results. Tackling the challenges also reveals that the arrival of maintenance demand congestion within a short period results in a high requirement for maintenance resources. As a result, tardiness as well as the high chance loss cost in rejecting aircraft maintenance requests may occur during the peak hours.

From the perspective of management, the negative impact of the lack of coordination of maintenance resources needs to be carefully handled, as the lack of coordination and low utilization of maintenance resources lead to a reduction of the client's satisfaction rate, as well as the company profit. Moreover, the arrival pattern of the demands should be carefully analyzed well in advance so as to prepare the arrangements for maintenance resources supply in real-world practice.

For the future research directions on this topic, it is possible to (1) consider advanced joint decision making among independent service companies and multiple airlines and understand the arrival pattern proactively, aiming to avoid the congestion of maintenance requests arriving within short period. Joint maintenance planning is able to improve the utilization of maintenance resources and service quality by the maintenance service provider, allowing the MRO service providers to have sufficient time to arrange and adjust their service capacity, especially manpower supply. When congested arrivals of maintenance demands occurs, joint decision-making allows the service provider and airline companies to adjust their respective maintenance plans among multiple parties in a flexible manner; (2) incorporate uncertainties in maintenance planning, including unscheduled maintenance and limited information regarding the arrival pattern of aircraft under uncertain scenarios (Zhang and Mahadevan, 2017; Leal de Matos and Powell, 2003; Kenan et al., 2018), so as to makes the optimization approach closer to real operations; (3) to further enhance the effectiveness and efficiency in solving two subproblems of the proposed two-stage optimization approach. It is possible to incorporate some enhanced search mechanisms according the model and problem nature. In addition, developing exact algorithms and improving heuristic algorithms based on the decomposition approaches proposed in this paper, given the effectiveness of the linkage constraints, is proposed in our two-stage optimization approach.

Credit authorship contribution statement

Yichen Qin: Conceptualization, Methodology, Software, Writing - original draft. **J.H. Zhang:** Supervision, Conceptualization, Methodology, Formal analysis, Writing - review & editing, Funding acquisition. **Felix T.S. Chan:** Visualization, Investigation, Software, Project administration. **S.H. Chung:** Data curation, Validation, Writing - review & editing. **B. Niu:** Data curation, Investigation. **T. Qu:** Data curation, Formal analysis.

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References

- Ng, K. K. H., Lee, C. K. M., Chan, F. T. S., & Lv, Y. (2018). Review on meta-heuristics approaches for airside operation research. *Applied Soft Computing*, 66, 104–133.
- Lee, C. K. M., Ng, K. K. H., Chan, H. K., Choy, K. L., Tai, W. C., & Choi, L. S. (2018). A multi-group analysis of social media engagement and loyalty constructs between full-service and low-cost carriers in Hong Kong. *Journal of Air Transport Management*, 73, 46–57.
- K. K. H. Ng, M. H. M. Tang, and C. K. M. Lee, "Design and development of a performance evaluation system for the aircraft maintenance industry," in 2015 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), Singapore, Singapore, 2015, pp. 564–568.
- Ng, K. K. H., Lee, C. K. M., Chan, F. T. S., Chen, C.-H., & Qin, Y. (2020). A two-stage robust optimisation for terminal traffic flow problem. *Applied Soft Computing*, 89, 106048.
- Ertogral, K., & Öztürk, F. S. (2019). An integrated production scheduling and workforce capacity planning model for the maintenance and repair operations in airline industry. *Computers & Industrial Engineering*, 127, 832–840.
- Ng, K. K. H., Lee, C. K. M., Chan, F. T. S., & Qin, Y. C. (2017). Robust aircraft sequencing and

- scheduling problem with arrival/departure delay using the min-max regret approach. *Transportation Research Part E-Logistics and Transportation Review*, 106, 115–136.
- K. K. H. Ng, C. K. M. Lee, and F. T. S. Chan, "A robust optimisation approach to the aircraft sequencing and scheduling problem with runway configuration planning," in 2017 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM), Singapore, Singapore, 2017, pp. 40–44.
- Wang, X. Y., Wang, C. S., & Heuzey, M. C. (2016). Complexation of chitosan and gelatin: From soluble complexes to colloidal gel. *International Journal of Polymeric Materials and Polymeric Biomaterials*, 65, 96–104.
- ITAT (2015). "Airline Maintenance Cost Executive. Commentary,".
- Eriksson, S., & Steenhuis, H. (2014). *The Global Commercial Aviation Industry*. New York: Taylor & Francis.
- Knotts, R. M. H. (1999). Civil aircraft maintenance and support fault diagnosis from a business perspective. *Journal of Quality in Maintenance Engineering*, 5, 335–348.
- Marcontelli, D. (2013). MRO's offshore edge shrinking. *Aviation Week & Space Technology*, 175, 56.
- Chen, G., He, W., Leung, L. C., Lan, T., & Han, Y. (2017). Assigning licenced technicians to maintenance tasks at aircraft maintenance base: A bi-objective approach and a Chinese airline application. *International Journal of Production Research*, 55, 5550–5563.
- Datta, P. P., Srivastava, A., & Roy, R. (2013). A simulation study on maintainer resource utilization of a fast jet aircraft maintenance line under availability contract. *Computers in Industry*, 64, 543–555.
- Qin, Y., Chan, F. T. S., Chung, S. H., & Qu, T. (2017). "Development of MILP model for integrated aircraft maintenance scheduling and multi-period parking layout planning problems," presented at the 2017 4th International Conference on Industrial Engineering and Applications. Japan: Nagoya.
- Qin, Y., Wang, Z. X., Chan, F. T. S., Chung, S. H., & Qu, T. (2019). A mathematical model and algorithms for the aircraft hangar maintenance scheduling problem. *Applied Mathematical Modelling*, 67, 491–509.
- Sriram, C., & Haghighi, A. (2003). An optimization model for aircraft maintenance scheduling and re-assignment. *Transportation Research Part a-Policy and Practice*, 37, 29–48.
- Clarke, L., Johnson, E., Nemhauser, G., & Zhu, Z. X. (1997). The aircraft rotation problem. *Annals of Operations Research*, 69, 33–46.
- Lee, S. G., Ma, Y. S., Thimm, G. L., & Verstraeten, J. (2008). Product lifecycle management in aviation maintenance, repair and overhaul. *Computers in Industry*, 59, 296–303.
- Zhu, H., Gao, J., Li, D., & Tang, D. (2012). A Web-based Product Service System for aerospace maintenance, repair and overhaul services. *Computers in Industry*, 63, 338–348.
- C. K. M. Lee, Y. Z. Huo, S. Z. Zhang, and K. K. H. Ng, "Design of a Smart Manufacturing System with the Application of Multi-access Edge Computing and Blockchain Technology," IEEE Access, pp. 1–1, 2020.
- Samaranayake, P., Lewis, G. S., Woxvold, E. R. A., & Toncich, D. (2002). Development of engineering structures for scheduling and control of aircraft maintenance. *International Journal of Operations & Production Management*, 22, 843–867.
- J. Van den Bergh, P. De Bruecker, J. Belien, and J. Peeters, "Aircraft maintenance operations: state of the art," FEB@Brussel research paper, 2013.
- Papachatzakis, N., Papachatzakis, P., Xanthakis, V., Mourtzis, D., & Chrysosouris, G. (2010). An approach to operational aircraft maintenance planning. *Decision Support Systems*, 48, 604–612.
- Qin, Y., Chan, F. T. S., Chung, S. H., Qu, T., & Niu, B. (2017). Aircraft parking stand allocation problem with safety consideration for independent hangar maintenance service providers. *Computers & Operations Research*.
- Hu, H., Ng, K. K. H., & Qin, Y. (2016). Robust Parallel Machine Scheduling Problem with Uncertainties and Sequence-Dependent Setup Time. *Scientific Programming*, 2016, 13.
- Belien, J., Demeulemeester, E., De Bruecker, P., Van den Bergh, J., & Cardoen, B. (2013). Integrated staffing and scheduling for an aircraft line maintenance problem. *Computers & Operations Research*, 40, 1023–1033.
- Eltoukhy, A. E. E., Wang, Z. X., Chan, F. T. S., & Chung, S. H. (2018). Joint optimization using a leader-follower Stackelberg game for coordinated configuration of stochastic operational aircraft maintenance routing and maintenance staffing. *Computers & Industrial Engineering*, 125, 46–68.
- Eltoukhy, A. E. E., Chan, F. T. S., Chung, S. H., & Niu, B. (2018). A model with a solution algorithm for the operational aircraft maintenance routing problem. *Computers & Industrial Engineering*, 120, 346–359.
- Diaz-Ramirez, J., Huertas, J. I., & Trigos, F. (2014). Aircraft maintenance, routing, and crew scheduling planning for airlines with a single fleet and a single maintenance and crew base. *Computers & Industrial Engineering*, 75, 68–78.
- Belien, J., Cardoen, B., & Demeulemeester, E. (2012). Improving Workforce Scheduling of Aircraft Line Maintenance at Sabena Technics. *Interfaces*, 42, 352–364.
- De Bruecker, P., Van den Bergh, J., Belien, J., & Demeulemeester, E. (2015). A model enhancement heuristic for building robust aircraft maintenance personnel rosters with stochastic constraints. *European Journal of Operational Research*, 246, 661–673.
- Eltoukhy, A. E. E., Wang, Z. X., Chan, F. T. S., & Fu, X. (2019). Data analytics in managing aircraft routing and maintenance staffing with price competition by a Stackelberg-Nash game model. *Transportation Research Part E: Logistics and Transportation Review*, 122, 143–168.
- Liang, Z., Feng, Y., Zhang, X. N., Wu, T., & Chaovallitwongse, W. A. (2015). Robust weekly aircraft maintenance routing problem and the extension to the tail assignment problem. *Transportation Research Part B-Methodological*, 78, 238–259.
- Gavranis, A., & Kozanidis, G. (2015). An exact solution algorithm for maximizing the fleet availability of a unit of aircraft subject to flight and maintenance requirements. *European Journal of Operational Research*, 242, 631–643.
- Feng, Q., Bi, W. J., Chen, Y. R., Ren, Y., & Yang, D. Z. (2017). Cooperative game approach based on agent learning for fleet maintenance oriented to mission reliability. *Computers & Industrial Engineering*, 112, 221–230.
- Lee, C. K., Zhang, S., & Ng, K. K. (2019). Design of An Integration Model for Air Cargo Transportation Network Design and Flight Route Selection. *Sustainability*, 11, 5197.
- Lee, C. K. M., Zhang, S., & Ng, K. K. H. (2019). In-plant logistics simulation model for the catering service industry towards sustainable development: A case study. *Sustainability*, 11, 3655.
- Yan, S. Y., Yang, T. H., & Chen, H. H. (2004). Airline short-term maintenance manpower supply planning. *Transportation Research Part a-Policy and Practice*, 38, 615–642.
- Zachariadis, E. E., Tarantilis, C. D., & Kiranoudis, C. T. (2016). The Vehicle Routing Problem with Simultaneous Pick-ups and Deliveries and Two-Dimensional Loading Constraints. *European Journal of Operational Research*, 251, 369–386.
- Ng, K. K. H., Lee, C. K. M., Zhang, S. Z., Wu, K., & Ho, W. (2017). A multiple colonies artificial bee colony algorithm for a capacitated vehicle routing problem and re-routing strategies under time-dependent traffic congestion. *Computers & Industrial Engineering*, 109, 151–168.
- Wei, L. J., Zhang, Z. Z., Zhang, D. F., & Lim, A. (2015). A variable neighborhood search for the capacitated vehicle routing problem with two-dimensional loading constraints. *European Journal of Operational Research*, 243, 798–814.
- Lee, C. K. M., Lin, B., Ng, K. K. H., Lv, Y., & Tai, W. C. (2019). Smart robotic mobile fulfillment system with dynamic conflict-free strategies considering cyber-physical integration. *Advanced Engineering Informatics*, 42, 100998.
- Cheang, B., Gao, X., Lim, A., Qin, H., & Zhu, W. B. (2012). Multiple pickup and delivery traveling salesman problem with last-in-first-out loading and distance constraints. *European Journal of Operational Research*, 223, 60–75.
- Cote, J. F., Gendreau, M., & Potvin, J. Y. (2014). An Exact Algorithm for the Two-Dimensional Orthogonal Packing Problem with Unloading Constraints. *Operations Research*, 62, 1126–1141.
- Cherkesly, M., Desaulniers, G., & Laporte, G. (2015). A population-based metaheuristic for the pickup and delivery problem with time windows and LIFO loading. *Computers & Operations Research*, 62, 23–35.
- Paes, F. G., Pessoa, A. A., & Vidal, T. (2017). A hybrid genetic algorithm with decomposition phases for the Unequal Area Facility Layout Problem. *European Journal of Operational Research*, 256, 742–756.
- Solimanpur, M., & Jafari, A. (2008). Optimal solution for the two-dimensional facility layout problem using a branch-and-bound algorithm. *Computers & Industrial Engineering*, 55, 606–619.
- Xie, W., & Sahinidis, N. V. (2008). A branch-and-bound algorithm for the continuous facility layout problem. *Computers & Chemical Engineering*, 32, 1016–1028.
- Anjos, M. F., & Vieira, M. V. C. (2017). Mathematical optimization approaches for facility layout problems: The state-of-the-art and future research directions. *European Journal of Operational Research*, 261, 1–16.
- Dunker, T., Radons, G., & Westkamper, E. (2005). Combining evolutionary computation and dynamic programming for solving a dynamic facility layout problem - Discrete optimization. *European Journal of Operational Research*, 165, 55–69.
- Xu, J. P., & Song, X. L. (2015). Multi-objective dynamic layout problem for temporary construction facilities with unequal-area departments under fuzzy random environment. *Knowledge-Based Systems*, 81, 30–45.
- Tyagi, S., Shukla, N., & Kulkarni, S. (2016). Optimal design of fixture layout in a multi-station assembly using highly optimized tolerance inspired heuristic. *Applied Mathematical Modelling*, 40, 6134–6147.
- Mohammadi, M., & Forghani, K. (2014). A novel approach for considering layout problem in cellular manufacturing systems with alternative processing routings and sub-contracting approach. *Applied Mathematical Modelling*, 38, 3624–3640.
- Bagheri, M., & Bashiri, M. (2014). A new mathematical model towards the integration of cell formation with operator assignment and inter-cell layout problems in a dynamic environment. *Applied Mathematical Modelling*, 38, 1237–1254.
- Ahmadi, A., & Jokat, M. R. A. (2016). An efficient multiple-stage mathematical programming method for advanced single and multi-floor facility layout problems. *Applied Mathematical Modelling*, 40, 5605–5620.
- Bozer, Y. A., & Rim, S. C. (1996). A branch and bound method for solving the bidirectional circular layout problem. *Applied Mathematical Modelling*, 20, 342–351.
- Bennell, J. A., & Oliveira, J. F. (2008). The geometry of nesting problems: A tutorial. *European Journal of Operational Research*, 184, 397–415.
- Bennell, J. A., & Oliveira, J. F. (2009). A tutorial in irregular shape packing problems. *Journal of the Operational Research Society*, 60, S93–S105.
- Alvarez-Valdes, R., Martinez, A., & Tamarit, J. M. (2013). A branch & bound algorithm for cutting and packing irregularly shaped pieces. *International Journal of Production Economics*, 145, 463–477.
- Fischetti, M., & Luzzi, I. (2009). Mixed-integer programming models for nesting problems. *Journal of Heuristics*, 15, 201–226.
- Martinez-Sykora, A., Alvarez-Valdes, R., Bennell, J., & Tamarit, J. M. (2015). Constructive procedures to solve 2-dimensional bin packing problems with irregular pieces and guillotine cuts. *Omega-International Journal of Management Science*, 52, 15–32.
- Cherri, L. H., Mundim, L. R., Andretta, M., Toledo, F. M. B., Oliveira, J. F., & Carravilla, M. A. (2016). Robust mixed-integer linear programming models for the irregular strip packing problem. *European Journal of Operational Research*, 253, 570–583.
- S. Zheng, Z. Yang, Z. He, N. Wang, C. Chu, and H. Yu, "Hybrid simulated annealing and reduced variable neighbourhood search for an aircraft scheduling and parking problem," International Journal of Production Research, pp. 1–21, 2019.
- Qin, Y., Chan, F. T. S., Chung, S. H., Qu, T., & Niu, B. (2018). Aircraft parking stand allocation problem with safety consideration for independent hangar maintenance service providers. *Computers & Operations Research*, 91, 225–236.
- Zhang, X., & Mahadevan, S. (2017). Aircraft re-routing optimization and performance assessment under uncertainty. *Decision Support Systems*, 96, 67–82.
- Leal de Matos, P. A., & Powell, P. L. (2003). Decision support for flight re-routing in Europe. *Decision Support Systems*, 34, 397–412.
- Kenan, N., Jebali, A., & Diabat, A. (2018). The integrated aircraft routing problem with optional flights and delay considerations. *Transportation Research Part E: Logistics and Transportation Review*, 118, 355–375.